

COMPETITION SECTION

Electric Charges and Field

GLIMPSES

1. **Electrostatics.** It is the study of electric charges at rest.
2. **Frictional electricity.** The property of rubbed substances due to which they attract light objects is called electricity. The electricity developed by rubbing or friction is called frictional or static electricity. The rubbed substances which show this property of attraction are said to be electrified or electrically charged substances.
3. **Electric charge.** It is an intrinsic property of elementary particles of matter which gives rise to electric force between various objects. It is a scalar quantity and its SI unit is coulomb (C).
4. **Positive and negative charges.** Benjamin Franklin introduced the present day convention that
 - (i) The charge developed on a glass rod when rubbed with silk is called positive charge.
 - (ii) The charge developed on a plastic/ebonite rod when rubbed with fur is called negative charge.
5. **Fundamental law of electrostatics.** Like charges repel and unlike charges attract each other.
6. **Electronic theory of frictional electricity.** During rubbing, electrons are transferred from one object to another. The object with excess of electrons develops a negative charge, while the object with deficit of electrons develops a positive charge.
7. **Electrostatic induction.** It is the phenomenon of temporary electrification of a conductor in which opposite charges appear at its closer end and similar charges appear at its farther end in the presence of a nearby charged body. An insulated conductor can be positively or negatively charged by induction.
8. **Electroscope.** A device used for detecting an electric charge and identifying its polarity is called electroscope.
9. **Three basic properties of electric charges.** These are : (i) quantization, (ii) additivity, and (iii) conservation.
10. **Additivity of electric charge.** This means that the total charge of a system is the algebraic sum of all the individual charges located at different points inside the system.
11. **Quantization of electric charge.** This means that the total charge (q) of a body is always an integral multiple of a basic quantum of charge (e) i.e.,
$$q = ne, \text{ where } n = 0, \pm 1, \pm 2, \pm 3, \dots$$
Faraday's laws of electrolysis and Millikan's oil drop experiment established the quantum nature of electric charge.
For macroscopically large charges, the quantization of charge can be ignored.
12. **Basic quantum of charge.** The smallest amount of charge or the basic quantum of charge is the charge on an electron or proton. Its exact magnitude is
$$e = 1.602182 \times 10^{-19} \text{ C}$$
13. **Law of conservation of charge.** It states that the total charge of a system remains unchanged with time. This means that when bodies are charged through friction, there is only transfer of charge from one body to another but no net creation or destruction of charge takes place.

14. **Coulomb's law.** The force of attraction or repulsion between two stationary point charges q_1 and q_2 is directly proportional to the product $q_1 q_2$ and inversely proportional to the square of the distance r between them. Mathematically,

$$F = k \frac{q_1 q_2}{r^2}$$

The proportionality constant k depends on the nature of the medium between the two charges and the system of units chosen to measure F, q_1, q_2 and r . For free space and in SI units,

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2},$$

ϵ_0 is called permittivity of free space and its value is $8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$.

Hence Coulomb's law in SI units may be expressed as

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}.$$

15. **SI unit of charge is coulomb (C).** It is that amount of charge that repels an equal and similar charge with a force of $9 \times 10^9 \text{ N}$ when placed in vacuum at a distance of one metre from it.
16. **Permittivity (ϵ).** It is the property of a medium which determines the electric force between two charges situated in that medium.
17. **Dielectric constant or relative permittivity.** The ratio (ϵ/ϵ_0) of the permittivity of the given medium to that of free space is known as relative permittivity (ϵ_r) or dielectric constant (κ) of the given medium,

$$\epsilon_r \text{ or } \kappa = \frac{\epsilon}{\epsilon_0} = \frac{F_{\text{vac}}}{F_{\text{med}}}$$

The dielectric constant of a medium may be defined as the ratio of the force between two charges placed some distance apart in free space to the force between the same two charges when they are placed the same distance apart in the given medium.

Coulomb's law for any medium other than vacuum can be written as

$$F_{\text{med}} = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0 \kappa} \frac{q_1 q_2}{r^2} = \frac{F_{\text{vac}}}{\kappa}.$$

18. **Electrostatic force vs. gravitational force.** Electrostatic forces are much stronger than

gravitational forces. The ratio of the electric force and gravitational force between a proton and an electron is

$$\frac{F_e}{F_g} = \frac{ke^2}{Gm_p m_e} \approx 2.27 \times 10^{39}$$

19. **Principle of superposition of electrostatic forces.**

When a number of charges are interacting, the total force on a given charge is the vector sum of the forces exerted on it due to all other charges. The force between two charges is not affected by the presence of other charges. The total force on charge q_1 due to the charges $q_2, q_3, \dots, q_N$ will be

$$\begin{aligned} \vec{F}_1 &= \vec{F}_{12} + \vec{F}_{13} + \dots + \vec{F}_{1N} \\ &= \frac{q_1}{4\pi\epsilon_0} \sum_{i=2}^N \frac{q_i}{r_{1i}^2} \hat{r}_{1i} \\ &= \frac{q_1}{4\pi\epsilon_0} \sum_{i=2}^N \frac{q_i (\vec{r}_1 - \vec{r}_i)}{|\vec{r}_1 - \vec{r}_i|^3} \end{aligned}$$

$$\text{where } \hat{r}_{1i} = \frac{\vec{r}_1 - \vec{r}_i}{|\vec{r}_1 - \vec{r}_i|}$$

= a unit vector pointing from q_i to q_1 .

20. **Electric field.** An electric field is said to exist at a point, if a force of electrical origin is exerted on a stationary charge placed at that point. Quantitatively, it is defined as the electrostatic force per unit test charge acting on a vanishingly small positive test charge placed at the given point. Mathematically,

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0}$$

Electric field is a vector quantity whose direction is same as that of the force exerted on a positive test charge.

21. **Units and dimensions of electric field.** The SI unit of electric field is newton per coulomb (NC^{-1}) or volt per metre (Vm^{-1}). The dimensions of electric field are

$$\begin{aligned} [E] &= \frac{\text{Force}}{\text{Charge}} = \frac{\text{MLT}^{-2}}{\text{C}} \\ &= \frac{\text{MLT}^{-2}}{\text{AT}} = [\text{MLT}^{-3}\text{A}^{-1}] \end{aligned}$$

22. **Electric field due to a point charge.** The electric field of a point charge q at distance r from it is given by

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

If q is positive, E points radially outwards and if q is negative, E points radially inwards. This field is spherically symmetric.

23. **Electric field due to a system of point charges : Superposition principle for electric fields.** The principle states that the electric field at any point due to a group of point charges is equal to the vector sum of the electric fields produced by each charge individually at that point, when all other charges are assumed to be absent.

$$\begin{aligned}\vec{E} &= \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_N \\ &= \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{r_{iP}^2} \hat{r}_{iP} \\ &= \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)\end{aligned}$$

24. **Continuous charge distribution.** When the charge involved is much greater than the charge on an electron, we can ignore its quantum nature and assume that the charge is distributed in a continuous manner. This is known as a continuous charge distribution.

$$\text{Volume charge density, } \rho = \frac{dq}{dV} \text{ Cm}^{-3}$$

$$\text{Surface charge density, } \sigma = \frac{dq}{dS} \text{ Cm}^{-2}$$

$$\text{Linear charge density, } \lambda = \frac{dq}{dL} \text{ Cm}^{-1}$$

25. **Electrostatic force and field due to a continuous charge distribution.** The total force on a charge q_0 due to a continuous charge distribution is given by

$$\vec{F}_{\text{cont}} = \vec{F}_V + \vec{F}_S + \vec{F}_L$$

$$\text{or } \vec{F}_{\text{cont}} = \frac{q_0}{4\pi\epsilon_0} \left[\int_V \frac{\rho}{r^2} \hat{r} dV + \int_S \frac{\sigma}{r^2} \hat{r} dS + \int_L \frac{\lambda}{r^2} \hat{r} dL \right]$$

$$\vec{E}_{\text{cont}} = \frac{\vec{F}_{\text{cont}}}{q_0}$$

$$= \frac{1}{4\pi\epsilon_0} \left[\int_V \frac{\rho}{r^2} \hat{r} dV + \int_S \frac{\sigma}{r^2} \hat{r} dS + \int_L \frac{\lambda}{r^2} \hat{r} dL \right]$$

26. **Electric field due to a general charge distribution.** It is given by

$$\begin{aligned}\vec{E}_{\text{total}} &= \vec{E}_{\text{discret}} + \vec{E}_{\text{cont}} \\ &= \frac{1}{4\pi\epsilon_0} \left[\sum_{i=1}^N \frac{q_i}{r_i^2} \hat{r}_i + \int_V \frac{\rho}{r^2} \hat{r} dV \right. \\ &\quad \left. + \int_S \frac{\sigma}{r^2} \hat{r} dS + \int_L \frac{\lambda}{r^2} \hat{r} dL \right]\end{aligned}$$

27. **Electric dipole and dipole moment.** An electric dipole is a pair of equal and opposite charges $+q$ and $-q$ separated by some distance $2a$. Its dipole moment is given by

$$\begin{aligned}\vec{p} &= \text{Either charge} \times \text{vector drawn from } -q \text{ to } +q \\ &= q \times 2a\end{aligned}$$

Magnitude of dipole moment, $p = q \times 2a$

Dipole moment is a vector quantity having direction along the dipole axis from $-q$ to $+q$. Its SI unit is coulomb metre (Cm).

28. **Electric field at an axial point of a dipole.** The dipole field on the axis at distance r from the centre is

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2pr}{(r^2 - a^2)^2} \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3} \text{ for } r \gg a.$$

At any axial point, the direction of dipole field is along the direction of dipole moment $\vec{p}$.

29. **Electric field at an equatorial point of a dipole.** The electric field at a point on the perpendicular bisector of the dipole at distance r from its centre is

$$E_{\text{equa}} = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{(r^2 + a^2)^{3/2}} \approx \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3} \text{ for } r \gg a.$$

At any equatorial point, the direction of dipole field is antiparallel to the direction of dipole moment $\vec{p}$.

In contrast to $1/r^2$ dependence of the electric field of a point charge, the dipole field has $1/r^3$ dependence. Moreover, the electric field due to a short dipole at a certain distance along the axis is twice the electric field at the same distance along the equatorial line.

30. **Torque on a dipole in a uniform electric field.** The torque on a dipole of moment p when placed

in a uniform electric field at an angle θ with it is given by

$$\tau = pE \sin \theta$$

In vector rotation, $\vec{\tau} = \vec{p} \times \vec{E}$

When the dipole is released, the torque $\vec{\tau}$ tends to align the dipole along the field $\vec{E}$.

If $E = 1$ unit and $\theta = 90^\circ$, then $\tau = p$. So dipole moment may also be defined as the torque acting on an electric dipole placed perpendicular to a uniform electric field of unit strength.

31. **Electric lines of force.** An electric line of force may be defined as the curve along which a small positive charge would tend to move when free to do so in an electric field and the tangent to which at any point gives the direction of electric field at that point.

32. **Important properties of electric lines of force.** These are :

- (i) Lines of force are continuous curves without any breaks.
- (ii) No two lines of force can cross each other.
- (iii) They start at positive charges and end at negative charges—they cannot form closed loops.
- (iv) The relative closeness of the lines of force indicates the strength of electric field at different points.
- (v) They are always normal to the surface of a conductor.
- (vi) They have a tendency to contract lengthwise and expand laterally.

33. **Electric flux.** The electric flux through a given area represents the total number of electric lines of force passing normally through that area. If the electric field $\vec{E}$ makes an angle θ with the normal to the area elements ΔS , then the electric flux is

$$\Delta\phi_E = E\Delta S \cos \theta = \vec{E} \cdot \Delta \vec{S}$$

The electric flux through any surface S , open or closed, is equal to the surface integral of $\vec{E}$ over the surface S .

$$\phi_E = \int_S \vec{E} \cdot d\vec{S}$$

Electric flux is a scalar quantity.
SI unit of electric flux = $\text{Nm}^2 \text{C}^{-1}$.

34. **Gaussian surface.** Any hypothetical closed surface enclosing a charge is called the Gaussian surface of that charge.

35. **Gauss's theorem.** The total flux of electric field $\vec{E}$ through a closed surface $\vec{S}$ is equal to $1/\epsilon_0$ times the charge q enclosed by the surface $\vec{S}$.

$$\phi_E = \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

36. **Electric field of a line charge.** The electric field of a long straight wire of uniform linear charge density λ ,

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad \text{i.e.,} \quad E \propto \frac{1}{r}$$

where r is the perpendicular distance of the wire from the observation point.

37. **Electric field of an infinite plane sheet of charge.** It does not depend on the distance of the observation point from the plane sheet.

$$E = \frac{\sigma}{2\epsilon_0}$$

where σ = uniform surface charge density.

38. **Electric field of two positively charged parallel plates.** If the two plates have surface charge densities σ_1 and σ_2 such that $\sigma_1 > \sigma_2 > 0$, then

$$E = \pm \frac{1}{2\epsilon_0} (\sigma_1 + \sigma_2) \quad (\text{Outside the plates})$$

$$E = \frac{1}{2\epsilon_0} (\sigma_1 - \sigma_2) \quad (\text{Inside the plates})$$

39. **Electric field of two equally and oppositely charged parallel plates.** If the two plates have surface charge densities $\pm \sigma$, then

$$E = 0 \quad (\text{For outside points})$$

$$E = \frac{\sigma}{\epsilon_0} \quad (\text{For inside points})$$

40. **Electric field of a thin spherical shell.** If R is the radius and σ , the surface charge density of the shell, then

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \quad \text{For } r > R \quad (\text{Outside points})$$

$$E = 0 \quad \text{For } r < R \quad (\text{Inside points})$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R^2} \quad \text{For } r = R \quad (\text{At the surface})$$

$$\text{where } q = 4\pi R^2 \sigma$$

41. **Electric field of a uniformly charged solid sphere.** If ρ is the uniform volume charge density and R radius of the sphere, then

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2} \quad \text{For } r > R \quad (\text{Outside points})$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{qr}{R^3} \quad \text{For } r < R \quad (\text{Inside points})$$

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R^2} \quad \text{For } r = R \quad (\text{At the surface})$$

$$\text{where } q = \frac{4}{3} \pi R^3 \rho$$

COMPETITION SECTION

Electrostatic Potential and Capacitance

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1. **Potential difference.** The potential difference between two points is defined as the amount of work done in bringing a unit positive charge from one point to another against the electrostatic forces.

$$\text{Potential difference} = \frac{\text{Work done}}{\text{Charge}}$$

or
$$V_{AB} = V_B - V_A = \frac{W_{AB}}{q}$$

2. **SI unit of potential difference is volt (V).** The potential difference between two points in an electric field is said to be 1 volt if 1 joule of work has to be done in moving a positive charge of 1 coulomb from one point to the other against the electrostatic forces.

$$1 \text{ V} = 1 \text{ J C}^{-1} = 1 \text{ N m C}^{-1}$$

3. **Electric potential.** It is defined as the amount of work done in bringing a unit positive charge from infinity to the observation point against the electrostatic forces.

$$\text{Electric potential} = \frac{\text{Work done}}{\text{Charge}}$$

or
$$V = \frac{W}{q}$$

Electric potential is a scalar quantity.

4. **SI unit of electric potential is volt.** The electric potential at a point in an electric field is said to be 1 volt if one joule of work has to be done in moving a positive charge of 1 coulomb from infinity to that point against the electrostatic forces.

5. **Electric potential due to a point charge.** The electric potential of a point charge q at distance r from it is given by

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r} \quad \text{i.e.,} \quad V \propto \frac{1}{r}$$

It is spherically symmetric.

6. **Electric potential due to a dipole.** Electric potential at a point having position vector $\vec{r}$, due to a dipole of moment $\vec{p}$ at the origin is given by

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{\vec{p} \cdot \vec{r}}{r^3} = \frac{1}{4\pi\epsilon_0} \cdot \frac{p \cos \theta}{r^2}$$

At points on the axial line of the dipole ($\theta = 0^\circ$ or 180°),

$$V_{\text{axial}} = \pm \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^2}$$

At points on the equatorial line of the dipole ($\theta = 90^\circ$),

$$V_{\text{equa}} = 0.$$

7. **Electric potential due to a group of N point charges.** If $r_1, r_2, r_3 \dots r_N$ are the distances of N point charges from the observation point, then

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \dots + \frac{q_N}{r_N} \right]$$
$$= \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{r_i}$$

8. **Determination of electric field from electric potential.** The rate of change of potential with distance is called potential gradient. Electric

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field at any point is equal to the negative of the potential gradient at that point

$$E = -\frac{dV}{dr}$$

∴ SI unit of electric field = Vm^{-1}

The direction of $\vec{E}$ is in the direction of steepest decrease of potential.

9. **Determination of electric potential from electric field.** The electric potential at a point having position vector $\vec{r}$ is given by

$$V = - \int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r}$$

10. **Equipotential surface.** Any surface that has same electric potential at every point on it is called an equipotential surface. The surface of a charged conductor is an equipotential surface. Some of the important properties of equipotential surface are as follows :

- (i) No work is done in moving a test charge over an equipotential surface.
 - (ii) Electric field is always normal to the equipotential surface at every point.
 - (iii) Equipotential surfaces are close together in the regions of strong field and farther apart in the regions of weak field.
 - (iv) No two equipotential surfaces can intersect each other.
11. **Electric potential energy.** The electric potential energy of a system of point charges is defined as the amount of work done in assembling the charges at their locations by bringing them in, from infinity.

P.E. of a charge = Charge $\times$ Electric potential at the given point

It is measured in joule (J) or electron volt (eV).

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

12. **Potential energy of a system of two point charges.** If two point charges q_1 and q_2 are separated by distance r_{12} , then their potential energy is

$$U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r_{12}}$$

13. **Potential energy of a system of three point charges.** It is given by

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_3 q_1}{r_{31}} \right]$$

14. **Potential energy of N point charges.** It is given by

$$U = \frac{1}{4\pi\epsilon_0} \sum_{\text{All pairs}} \frac{q_i q_j}{r_{ij}}$$

$$= \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \sum_{\substack{j=1 \\ i \neq j}}^N \frac{q_i q_j}{r_{ij}}$$

15. **Potential energy of a dipole in a uniform electric field.** It is equal to the amount of work done in turning the dipole from orientation θ_1 to θ_2 in the field E .

$$U = -pE(\cos \theta_2 - \cos \theta_1)$$

If initially the dipole is perpendicular to the field E , $\theta_1 = 90^\circ$ and $\theta_2 = \theta$ (say), then

$$U = -pE \cos \theta = -\vec{p} \cdot \vec{E}$$

When $\theta = 0^\circ$, $U = -pE$ i.e., the potential energy of the dipole is minimum. The dipole is in *stable equilibrium*.

When $\theta = 90^\circ$, $U = 0$

When $\theta = 180^\circ$, $U = +pE$

i.e., the potential energy of the dipole is maximum. The dipole is in *unstable equilibrium*.

16. **Conductors and insulators.** Conductors are the substances which allow large scale physical movement of electric charges through them when an external electric field is applied. They contain a large number of free electrons. Insulators are the substances which do not allow physical movement of electric charges through them when an external electric field is applied. They contain a negligibly small number of free charge carriers.

17. **Electrostatic properties of a conductor.** When placed in an electrostatic field, a conductor shows the following properties :

- (i) Net electrostatic field is zero in the interior of a conductor.
- (ii) Just outside the surface of a conductor, electric field is normal to the surface.

- (iii) The net charge in the interior of a conductor is zero and any excess charge resides on its surface.
- (iv) Potential is constant within and on the surface of a conductor.
- (v) Electric field at the surface of a charged conductor is proportional to the surface charge density.
- (vi) Electric field is zero in the cavity of a hollow charged conductor.

18. **Electrostatic shielding.** The phenomenon of making a region free from any electric field is called electrostatic shielding. It is based on the fact that electric field vanishes inside the cavity of a hollow conductor.

19. **Capacitance of a conductor.** It is the charge required to increase the potential of a conductor by unit amount.

$$\text{Capacitance} = \frac{\text{Charge}}{\text{Potential}}$$

$$\text{or } C = \frac{q}{V}$$

20. **Capacitance of a spherical conductor.** It is proportional to the radius R of the spherical conductor.

$$C = 4\pi\epsilon_0 R$$

21. **Capacitor.** It is an arrangement of two conductors separated by an insulating medium that is used to store electric charge and electric energy.

22. **Capacitance of a capacitor.** The capacitance of a capacitor is the charge required to be supplied to one of its conductors so as to increase the potential difference between two conductors by unit amount.

$$C = \frac{q}{V}$$

23. **Farad.** It is the SI unit of capacitance. The capacitance of a capacitor is 1 farad (F) if 1 coulomb of charge is transferred from its one plate to another on applying a potential difference of 1 volt across the two plates.

$$1 \text{ farad} = \frac{1 \text{ coulomb}}{1 \text{ volt}} \quad \text{or} \quad 1 \text{ F} = \frac{1 \text{ C}}{1 \text{ V}}$$

$$1 \text{ mF} = 10^{-3} \text{ F}, 1 \mu\text{F} = 10^{-6} \text{ F}, 1 \text{ pF} = 10^{-12} \text{ F}.$$

24. **Parallel plate capacitor.** It consists of two large parallel conducting plates, each of area A , and separated by a small distance d . Its capacitance is

$$C = \frac{\epsilon_0 A}{d}$$

25. **Spherical capacitor.** It consists of two concentric spherical conducting shells of inner and outer radii a and b .

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

26. **Cylindrical capacitor.** It consists of two coaxial conducting cylinders of inner and outer radii a and b and of common length l .

$$C = 2\pi\epsilon_0 \frac{l}{\log_e \frac{b}{a}} = 2\pi\epsilon_0 \frac{l}{2.303 \log_{10} \frac{b}{a}}$$

27. **Capacitors in series.** The equivalent capacitance C_s of number of capacitors connected in series is given by

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

In a series combination of capacitors, the charge on each capacitor is same but the potential difference across any capacitor is inversely proportional to its capacitance.

28. **Capacitors in parallel.** The equivalent capacitance of a number of capacitors connected in parallel is given by

$$C_p = C_1 + C_2 + C_3 + \dots$$

In a parallel combination of capacitors, the potential difference across each capacitor is same but the charge on each capacitor is proportional to its capacitance.

29. **Energy stored in a capacitor.** The energy stored in a capacitor of capacitance C and charge q with voltage V is

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \cdot \frac{Q^2}{C} = \frac{1}{2} QV$$

30. **Energy density.** The electrical energy stored per unit volume or energy density in a region with electric field E is

$$u = \frac{1}{2} \epsilon_0 E^2$$

31. **Common potential.** If a number of conductors of capacitances $C_1, C_2, C_3, \dots$, at potentials $V_1, V_2, V_3, \dots$, having charges $q_1, q_2, q_3, \dots$ respectively are placed in contact, their common potential V is given by

$$V = \frac{\text{Total charge}}{\text{Total capacitance}} = \frac{q_1 + q_2 + q_3 + \dots}{C_1 + C_2 + C_3 + \dots} \\ = \frac{C_1 V_1 + C_2 V_2 + C_3 V_3 + \dots}{C_1 + C_2 + C_3 + \dots}$$

32. **Loss of energy on sharing charges.** If two conductors of capacitances C_1 and C_2 at potentials V_1 and V_2 respectively are connected together, a loss of energy takes place which is given by

$$\Delta U = \frac{1}{2} \frac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2.$$

33. **Dielectric.** A dielectric is a substance which does not allow the flow of charges through it but permits them to exert electrostatic forces on one another. It is essentially an insulator which can be polarised through small localised displacements of its charges.

34. **Polar and non-polar dielectrics.** The dielectrics made of polar molecules (such as HCl , NH_3 , H_2O , CH_3OH , etc.) are called polar dielectrics. In a polar molecule, the centre of mass of positive charges does not coincide with the centre of mass of negative charges.

The dielectrics made of non-polar molecules are called non-polar dielectrics. In a non-polar molecule, the centre of mass of positive charges coincides with the centre of mass of negative charges e.g., H_2 , O_2 , CO_2 , CH_4 , etc.

35. **Polarisation of dielectric.** If the medium between the plates of a capacitor is filled with a dielectric, the electric field due to the charged plates induces a net dipole moment in the dielectric. This effect is called polarisation which induces a field in the opposite direction. The net electric field inside the dielectric and hence the potential difference between the plates are reduced. Consequently, the capacitance C increases from its value C_0 when there is vacuum.

$$C = \kappa C_0.$$

36. **Dielectric constant.** It is the ratio of the capacitance (C) of the capacitor with the

dielectric as the medium to its capacitance (C_0) when conductors are in vacuum.

$$\kappa = \frac{C}{C_0}$$

It is also equal to the ratio of the applied electric field (E_0) to the reduced value of electric field (E) on inserting the dielectric slab between the plates of the capacitor.

$$\kappa = \frac{E_0}{E} = \frac{E_0}{E_0 - E'}.$$

Here E' is the field set up due to polarisation of the dielectric in the opposite direction of E_0 .

37. **Capacitance of a parallel plate capacitor filled with a dielectric.**

$$C = \kappa C_0 = \frac{\epsilon_0 \kappa A}{d}.$$

38. **Capacitance of a parallel plate capacitor with a dielectric slab between its plates.** If t is the thickness of the dielectric slab and $t < d$, then

$$C = \frac{\epsilon_0 A}{d - t \left(1 - \frac{1}{\kappa} \right)}.$$

39. **Capacitance of a parallel plate capacitor with conducting slab between its plates.** For $t < d$,

$$C = \left(\frac{d}{d-t} \right) \frac{\epsilon_0 A}{d} = \left(\frac{d}{d-t} \right) C_0.$$

40. **Capacitance of a spherical capacitor filled with a dielectric.**

$$C = 4 \pi \epsilon_0 \kappa \frac{ab}{b-a}.$$

41. **Capacitance of a cylindrical capacitor filled with a dielectric**

$$C = \frac{2 \pi \epsilon_0 \kappa l}{2.303 \log_{10} \frac{b}{a}}.$$

42. **Van de Graaff generator.** It is an electrostatic generator capable of building up high potential differences of the order of 10^7 volts. It is based on the principle that when a charged conductor is brought into internal contact with a hollow conductor, it transfers whole of its charge to the hollow conductor, howsoever high the potential of the latter may be. Also, it uses discharging action of sharp points. It is used for accelerating charged particles.

COMPETITION SECTION

Current Electricity

GLIMPSES

1. **Current electricity.** The study of electric charges in motion is called current electricity.
2. **Electric current.** The flow of electric charges through a conductor constitutes electric current. Quantitatively, electric current across an area held perpendicular to the direction of flow of charge is defined as the amount of charge flowing across that area per unit time.

For a steady flow of charge, $I = \frac{Q}{t}$

If the rate of flow of charge varies with time, then

$$I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

SI unit of current is *ampere* (A).

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{1 \text{ second}} \quad \text{or} \quad 1 \text{ A} = 1 \text{ Cs}^{-1}$$

3. **Conventional and electronic currents.** The direction of motion of positive charges is taken as the direction of conventional current. Electrons being negatively charged, so the direction of electronic current is opposite to that of the conventional current.
4. **Electric current is a scalar quantity.** Although we represent current with an arrow, yet it is a scalar quantity. Electric currents do not obey the laws of vector addition.
5. **Electromotive force (emf).** The emf of a source may be defined as the work done by the source in taking a unit positive charge from its lower potential terminal to the higher potential terminal. Or, it is the energy supplied by the source in taking a unit positive charge once

round the complete circuit. It is equal to the terminal p.d. measured in open circuit.

$$\text{EMF} = \frac{\text{Work done}}{\text{Charge}} \quad \text{or} \quad \mathcal{E} = \frac{W}{q}$$

6. **SI unit of emf is volt.** If an electrochemical cell supplies energy of 1 joule for the flow of 1 coulomb of charge through the whole circuit (including the cell), then its emf is said to be one volt.
7. **Ohm's law.** The current flowing through a conductor is directly proportional to the potential difference across its ends, provided the temperature and other physical conditions remain unchanged.

$$V \propto I \quad \text{or} \quad V = RI \quad \text{or} \quad \frac{V}{I} = R$$

Here R is called the resistance of the conductor.

8. **Resistance.** It is the property by virtue of which a conductor opposes the flow of charges through it. It is equal to the ratio of the potential difference applied across the conductor to the current flowing through it. It depends on the length l and area of cross-section A of the conductor through the relation :

$$R = \rho \frac{l}{A}, \quad \rho = \text{resistivity of the material.}$$

9. **SI unit of resistance is ohm (Ω).** The resistance of a conductor is 1 ohm if a current of 1 ampere flows through it on applying a potential difference of 1 volt across its ends.

$$1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}} \quad \text{or} \quad 1 \Omega = 1 \text{ VA}^{-1}$$

10. **Resistivity or specific resistance.** It is the resistance offered by a unit cube of the material of a conductor.

$$\rho = \frac{RA}{l}$$

SI unit of $\rho = \Omega \text{ m}$

It depends on the nature of the material of the conductor and the physical conditions like temperature, pressure, etc.

11. **Current density.** It is the amount of charge flowing per second per unit area normal to the flow of charge. It is a vector quantity having the same direction as that of the motion of the positive charge.

For normal flow of charge,

$$j = \frac{q/t}{A} = \frac{I}{A}$$

In general, $I = jA \cos \theta = \vec{j} \cdot \vec{A}$

SI unit of current density = Am^{-2} .

12. **Conductance.** It is the reciprocal of resistance.

$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

or
$$G = \frac{1}{R}$$

SI unit of conductance = $\text{ohm}^{-1} = \text{mho}$
= siemen (S).

13. **Conductivity or specific conductance.** It is the reciprocal of resistivity.

$$\text{Conductivity} = \frac{1}{\text{Resistivity}}$$

or
$$\sigma = \frac{1}{\rho}$$

SI unit of conductivity

$$= \text{ohm}^{-1} \text{ m}^{-1} = \text{mho m}^{-1} = \text{S m}^{-1}$$

14. **Resistivities of different substances.** Metals have low resistivities in the range of $10^{-8} \Omega \text{ m}$ to $10^{-6} \Omega \text{ m}$. Insulators have resistivities more than $10^4 \Omega \text{ m}$. Semiconductors have intermediate resistivities lying between $10^{-6} \Omega \text{ m}$ to $10^4 \Omega \text{ m}$.

15. **Colour code for carbon resistors.**

Colour	Number	Multiplier
Black	0	10^0
Brown	1	10^1
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Grey	8	10^8
White	9	10^9

How to remember colour code :

B	B	R	O	Y	of	Great	Britain	had	Very	Good	Wife
↓	↓	↓	↓	↓		↓			↓	↓	↓
0	1	2	3	4		5	6		7	8	9

Tolerance :

Gold	Silver	No colour
$\pm 5\%$	$\pm 10\%$	$\pm 20\%$

A set of coloured co-axial bands is printed on the resistor which reveals the following facts :

- (1) The first band indicates the *first significant figure*.
 - (2) The second band indicates the *second significant figure*.
 - (3) The third band indicates the *power of ten* with which the above two significant figures must be multiplied to get the resistance value in ohms.
 - (4) The last band indicates the tolerance in per cent of the indicated value.
16. **Carriers of current.** In metals, free electrons are the charge carriers. In ionised gases, electrons and positively charged ions are the charge carriers. In an electrolyte, both positive and negative ions are the charge carriers. In semiconductors like Ge and Si, conduction is due to electrons and holes. A hole is a vacant state from which an electron has been removed and acts as a positive charge carrier.
17. **Drift velocity and relaxation time.** The average velocity acquired by the free electrons of a conductor in the opposite direction of the externally applied electric field is called drift velocity (v_d). The average time that elapses between the two

successive collisions of an electron is called relaxation time (τ).

$$v_d = \frac{eE\tau}{m} ; R = \frac{ml}{ne^2\tau A} ; \rho = \frac{m}{ne^2\tau}$$

$$I = enAv_d ; j = env_d$$

Here n = no. of free electrons per unit volume or free electron density and m = mass of an electron.

18. **Other forms of Ohm's law.** In terms of vector quantities like current density $\vec{j}$ and electric field $\vec{E}$, Ohm's law may be expressed as

$$\vec{j} = \sigma \vec{E} \quad \text{or} \quad \vec{E} = \rho \vec{j}$$

The equation $\vec{E} = \rho \vec{j}$ leads to another statement of Ohm's law i.e., a conducting material obeys Ohm's law when the resistivity of the material does not depend on the magnitude and direction of the applied electric field.

19. **Temperature coefficient of resistance (α).** It is defined as the change in resistance per unit original resistance per degree rise in temperature. It is given by

$$\alpha = \frac{R_2 - R_1}{R_1(t_2 - t_1)}$$

If $t_1 = 0^\circ\text{C}$ and $t_2 = t^\circ\text{C}$, then

$$\alpha = \frac{R_t - R_0}{R_0 \times t} \quad \text{or} \quad R_t = R_0(1 + \alpha t)$$

The unit of α is $^\circ\text{C}^{-1}$ or K^{-1} .

20. **Effect of temperature on resistance.** For metals α is positive i.e., resistance of metals increases with the increase in temperature.

For semiconductors and insulators, α is negative i.e., their resistance decreases with the increase in temperature.

For alloys like constantan and manganin, the temperature coefficient of resistance α is very small. So they are used for making standard resistors.

21. **Mobility of a charge carrier.** The mobility of a charge carrier is the drift velocity acquired by it in a unit electric field. It is given by

$$\mu = \frac{v_d}{E} = \frac{q\tau}{m}$$

For an electron, $\mu_e = \frac{e\tau_e}{m_e}$

For a hole, $\mu_h = \frac{e\tau_h}{m_h}$

SI unit of mobility = $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$

Practical unit of mobility = $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$.

22. **Relation between electric current and mobility.** For a conductor,

$$I = enA\mu_e E$$

For a semiconductor,

$$I = eAE(n\mu_e + p\mu_h)$$

and

$$\sigma = e(n\mu_e + p\mu_h)$$

where n and p are the electron and hole densities.

23. **Ohmic conductors.** The conductors which obey Ohm's law are called Ohmic conductors. For these conductors, V - I graph is a straight line passing through the origin. For example, a metallic conductor for small currents is an Ohmic conductor.

24. **Non-ohmic conductors.** The conductors which do not obey Ohm's law are called non-ohmic conductors. The non-ohmic situations may be of the following types :

- The straight line V - I graph does not pass through the origin.
- V - I relationship is non-linear.
- V - I relationship depends on the sign of V for the same absolute value of V .
- V - I relationship is non-unique.

Examples of non-ohmic conductors are water voltameter, thyristor, a p - n junction, etc.

25. **Superconductivity.** The phenomenon of complete loss of resistivity by certain metals and alloys when they are cooled below a certain temperature is called superconductivity. The temperature at which a substance undergoes a transition from normal conductor to superconductor in a zero magnetic field is called transition or critical temperature (T_c).

26. **Meissner effect.** The expulsion of the magnetic flux from a superconducting material when it is cooled to a temperature below the critical temperature in a magnetic field is called Meissner effect.

- 27... **Resistances in series.** When a number of resistances are connected in series, their equivalent resistance (R_s) is equal to the sum of the individual resistances.

$$R_s = R_1 + R_2 + R_3 + \dots$$

28. **Resistances in parallel.** When a number of resistances are connected in parallel, the reciprocal of their equivalent resistance (R_p) is equal to the sum of the reciprocals of the individual resistances.

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$$

For two resistances in parallel,

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

29. **Division of current in resistors joined in parallel.** The current is divided in resistors, connected in parallel, in the inverse ratio of their resistances.

$$I_1 = \frac{R_2}{R_1 + R_2} \cdot I$$

$$I_2 = \frac{R_1}{R_1 + R_2} \cdot I$$

30. **Internal resistance (r).** The resistance offered by the electrolyte of a cell to the flow of current between its electrodes is called internal resistance of the cell. It depends on (i) nature of the electrolyte, (ii) concentration of the electrolyte, (iii) distance between the electrodes, (iv) common area of the electrodes dipped in the electrolyte and (v) temperature of the electrolyte.

31. **Relations between emf ($\mathcal{E}$), terminal potential difference (V) and internal resistance (r).** The potential drop across the terminals of a cell when a current is being drawn from it is called its terminal potential difference. It is less than the emf of the cell in a closed circuit.

$$\mathcal{E} = V + Ir; \quad V = \mathcal{E} - Ir; \quad V = IR = \frac{\mathcal{E} R}{R + r}$$

$$r = \frac{\mathcal{E} - V}{I} = R \left[\frac{\mathcal{E} - V}{V} \right]; \quad I = \frac{\mathcal{E}}{R + r}; \quad I_{\max} = \frac{\mathcal{E}}{r}$$

Terminal p.d. of a cell when it is being charged is

$$V = \mathcal{E} + Ir, \quad V > \mathcal{E}.$$

32. **Cells in series.** If n cells of emf $\mathcal{E}$ and internal resistance r each are connected in series, then current drawn through external resistance R is

$$I = \frac{n\mathcal{E}}{R + nr}$$

33. **Cells in parallel.** If m cells are connected in parallel, then current drawn through external resistance R is

$$I = \frac{m\mathcal{E}}{mR + r}$$

34. **Cells in mixed grouping.** If n cells are connected in series in each row and m such rows are connected in parallel, then current drawn through an external resistance R is

$$I = \frac{mn\mathcal{E}}{mR + nr}$$

For maximum current, the external resistance must be equal to the total internal resistance, i.e.,

$$R = \frac{nr}{m}$$

or $mR = nr$.

35. **Heating effect of current.** The phenomenon of the production of heat in a resistor by the flow of an electric current through it is called heating effect of current or Joule heating. It is an irreversible process.

36. **Joule's law of heating.** It states that the amount of heat H produced in a resistor is

- (i) directly proportional to the square of current for a given R ,
- (ii) directly proportional to the resistance R for a given I ,
- (iii) inversely proportional to the resistance R for a given V , and
- (iv) directly proportional to the time t for which the current flows through the resistor.

Mathematically, this law can be expressed as

$$H = VIt \text{ joule}$$

$$= I^2 R t \text{ joule} = \frac{V^2 t}{R} \text{ joule}$$

$$\text{or } H = \frac{VIt}{4.18} \text{ cal}$$

$$= \frac{I^2 R t}{4.18} \text{ cal} = \frac{V^2 t}{4.18 R} \text{ cal}$$

37. **Electric power.** It is the rate at which an electric appliance converts electric energy into other forms of energy. Or, it is the rate at which work is done by a source of emf in maintaining an electric current through a circuit.

Electric power,

$$P = \frac{W}{t} = VI = I^2 R = \frac{V^2}{R}$$

38. **SI unit of power is watt.** The power of an appliance is one watt if one ampere of current flows through it on applying a potential difference of 1 volt across it.

$$1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}}$$

$$= 1 \text{ volt} \times 1 \text{ ampere}$$

or $1 \text{ W} = 1 \text{ Js}^{-1} = 1 \text{ VA}$

1 kilowatt (kW) = 1000 W.

39. **Electric energy.** It is the total work done in maintaining an electric current in an electric circuit for a given time.

$$\text{Electric energy} = \text{Electric power} \times \text{time}$$

$$W = Pt$$

$$= VI t \text{ joule} = I^2 R t \text{ joule}$$

40. **Units of electric energy.** The commercial unit of electric energy is kilowatt-hour (kWh) or Board of Trade (B.O.T.) unit. It is the electric energy consumed by an appliance of power 1000 watt in one hour.

$$1 \text{ kWh} = 1000 \text{ Wh}$$

$$= 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$$

Another unit of electric energy is watt hour.

$$1 \text{ watt hour} = 1 \text{ watt} \times 1 \text{ hour}$$

$$= 3.6 \times 10^3 \text{ J.}$$

41. **Power rating.** The power rating of an electrical appliance is the electrical energy consumed per second by the appliance when connected across the marked voltage of the mains.

$$P = \frac{V^2}{R} = I^2 R = VI.$$

42. **Power consumed by a series combination of appliances.** The reciprocal of the effective

power is equal to the sum of the reciprocals of the individual powers of the appliances which have been manufactured for working on the same voltage.

$$\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} + \dots$$

43. **Power consumed by a parallel combination of appliances.** The effective power is equal to the sum of the powers of the individual appliances.

$$P = P_1 + P_2 + P_3 + \dots$$

44. **Efficiency of a source of emf.** It is the ratio of the output power to the input power. If a source of emf $\mathcal{E}$ and internal resistance r is connected to an external resistance, then its efficiency will be

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

$$= \frac{VI}{\mathcal{E}I} = \frac{V}{\mathcal{E}} = \frac{R}{R+r}.$$

45. **Maximum Power Theorem.** It states that the output power of a source of emf is maximum when the external resistance in the circuit is equal to internal resistance of the circuit i.e., when $R = r$.

$$P_{\max} = \frac{\mathcal{E}^2}{4r}.$$

The efficiency of a source of emf is 50% when it delivers maximum power.

46. **Efficiency of an electric device.** It is the ratio of the output power to the input power.

$$\eta = \frac{\text{Output power}}{\text{Input power}}$$

(i) For an electric motor,

$$\eta = \frac{\text{Output mechanical power}}{\text{Input electric power}}$$

(ii) Input electric power

$$= \text{Output mechanical power}$$

$$+ \text{Power lost as heat}$$

The power output of an electric motor is maximum when its back emf is one-half the source emf, provided the resistance of the windings of the motor is negligible.

47. **Kirchhoff's laws.** These laws enable us to determine the currents and voltages in different parts of the electrical circuits.

First law or junction rule. In an electric circuit, the algebraic sum of currents at any junction is zero. Or, the sum of currents entering a junction is equal to the sum of the currents leaving that junction.

Mathematically,

$$\Sigma I = 0$$

Justification. This law is based on the law of conservation of charge. When the currents in a circuit are steady, charges cannot accumulate or originate at any point of the circuit.

Second law or loop rule. Around any loop of a network, the sum of changes in potential must be zero. Or, the algebraic sum of the emfs in any loop of a circuit is equal to the sum of the products of currents and resistances in it.

$$\Sigma \Delta V = 0$$

or $\Sigma \mathcal{E} = \Sigma IR$

Justification. This law is based on the law of conservation of energy. As the electrostatic force is a conservative force, the total work done by it along any closed path must be zero.

48. **Galvanometer.** It is a sensitive device to detect current in a circuit. It produces a deflection proportional to the electric current flowing through it.
49. **Potentiometer.** It is a device used to compare emfs of two sources. Its working is based on the principle that when a constant current flows through a wire of uniform cross-sectional area and composition, the p.d. across any length of the wire is directly proportional to that length.

$$V \propto l$$

or $V = kl$

where k is the potential drop per unit length which is called potential gradient. Potentiometer has two main uses.

- (i) **To compare the emfs of two cells.** If l_1 and l_2 are the balancing lengths of the potentiometer wire for the cells of emfs $\mathcal{E}_1$ and $\mathcal{E}_2$ respectively, then

$$\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{l_2}{l_1}$$

- (ii) **To find the internal resistance r of a cell.** If l_1 is the balancing length of the potentiometer wire without shunt and l_2 the balancing length with shunt R across the cell, then internal resistance of the cell will be

$$r = \frac{\mathcal{E} - V}{V} \times R = \frac{l_1 - l_2}{l_2} \times R$$

50. **Wheatstone bridge.** It is an arrangement of four resistances P , Q , R and S joined to form a quadrilateral $ABCD$ with a battery between A and C and a sensitive galvanometer between B and D . The resistances are so adjusted that no current flows through the galvanometer. The bridge is then said to be balanced. In the balanced condition,

$$\frac{P}{Q} = \frac{R}{S}$$

Knowing any three resistances, the fourth resistance can be computed. A wheatstone bridge is most sensitive when the resistances in its four arms are of the same order.

51. **Slide wire bridge or metre bridge.** It is an application of wheatstone bridge in which R is fixed and a balance point is obtained by varying P and Q i.e., by adjusting the position of a jockey on a 100 cm long resistance wire stretched between two terminals. If the balance point is obtained at length l , then

$$\frac{P}{Q} = \frac{R}{S} = \frac{l}{100 - l}$$

or $S = \left(\frac{100 - l}{l} \right) R$

Resistivity, $\rho = \frac{SA}{l}$
 $= \frac{S \times \pi r^2}{l}$

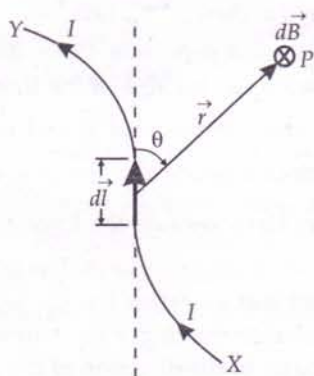
COMPETITION SECTION

Magnetic Effect of Current

GLIMPSES

1. **Oersted observation.** A compass needle suffers a deflection when it is placed near a wire carrying an electric current. When the direction of current is reversed, the direction of deflection of the needle also reverses. This conclusively proves that a current carrying conductor produces a magnetic field around it. This is called *magnetic effect of current*.
2. **Biot-Savart law.** According to this law, the magnetic field due to a current carrying element $\vec{dl}$ carrying current I at a point P at distance r from it is given by

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{I dl \sin \theta}{r^2}$$



In vector notation, $\vec{dB} = \frac{\mu_0 I}{4\pi} \cdot \frac{\vec{dl} \times \vec{r}}{r^3}$

Here θ is the angle between $\vec{dl}$ and $\vec{r}$ and $\mu_0 = 4\pi \times 10^{-7} \text{ Tm A}^{-1}$, is the permeability of

free space. The direction of $\vec{dB}$ is same as that of $\vec{dl} \times \vec{r}$ and is given by right-hand screw rule.

3. **Magnetic field due to straight current carrying conductor.** The magnetic field at a point at perpendicular distance 'a' from a straight conductor carrying current I is given by

$$B = \frac{\mu_0 I}{4\pi a} (\sin \phi_1 + \sin \phi_2)$$

where ϕ_1 and ϕ_2 are the angles which the perpendicular from the observation point to the conductor makes with the lines joining the ends of the conductor to the observation point.

The magnetic field due to a straight conductor of infinite length ($\phi_1 = \phi_2 = \frac{\pi}{2}$) is given by

$$B = \frac{\mu_0 I}{2\pi a}$$

4. **Rules for finding the direction of magnetic field due to straight current carrying conductor.**
 - (i) **Right hand thumb rule.** If we hold the straight conductor in the grip of our right hand in such a way that the extended thumb points in the direction of current, then the direction of curl of fingers will give the direction of the magnetic field.
 - (ii) **Maxwell's cork screw rule.** If a right handed screw be rotated along a wire so that it advances in the direction of current, then the direction in which the thumb rotates gives the direction of the magnetic field.

5. **Magnetic field of a circular current loop.** The magnetic field of a circular current loop of radius a carrying current I is

$$(i) \text{ At the centre of the loop : } B = \frac{\mu_0 I}{2a}$$

- (ii) At an axial point at distance r from the centre :

$$B = \frac{\mu_0 I a^2}{2(r^2 + a^2)^{3/2}}$$

6. **Rules for finding the direction of magnetic field due to circular current loop.**

(i) **Right hand thumb rule.** If we curl the fingers of our right hand around the circular wire with the fingers pointing in the direction of the current, then the extended thumb gives the direction of the magnetic field.

(ii) **Clock rule.** This rule gives the polarity of any face of the circular current loop. If the current round any face of the coil is in anticlockwise direction, it behaves like a north pole. If the current flows in the clockwise direction, it behaves like a south pole.

7. **Ampere's circuital law.** The line integral of the magnetic field $\vec{B}$ around any closed path is equal to μ_0 times the total current I threading the closed path.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

In a simplified form, the law states that if field $\vec{B}$ is directed along the tangent to every point on the perimeter L of a closed curve and its magnitude is constant along the curve, then

$$BL = \mu_0 I.$$

8. **Magnetic field of a straight solenoid.** A solenoid is a long insulated wire wound in the form of a helix. Its length is very large as compared to its diameter. The magnetic field of a straight solenoid carrying current I and having n turns per unit length is given by

$$(i) B = \mu_0 n I$$

(At a point well inside the solenoid)

$$(ii) B_{\text{end}} = \frac{1}{2} \mu_0 n I$$

(At either end of the solenoid)

9. **Magnetic field of a toroidal solenoid.** A solenoid bent into the form of a closed ring is called a

toroidal solenoid. The magnetic field inside the toroidal solenoid has a constant magnitude and tangential direction. It is given by

$$B = \mu_0 n I$$

where I is the current in the windings and n is the number of turns per unit length. The field lines are concentric circles.

10. **Force on a charge moving in a magnetic field.** A charge q moving with velocity $\vec{v}$ at an angle θ with the magnetic field $\vec{B}$ experiences the magnetic Lorentz force,

$$F = qvB \sin \theta$$

In vector notation, $\vec{F} = q(\vec{v} \times \vec{B})$

The direction of this force is perpendicular to both $\vec{v}$ and $\vec{B}$, and work done by it is zero. This force is maximum when the charged particle moves perpendicular to the direction of the field ($\theta = 90^\circ$) and minimum when the charged particle moves along the field ($\theta = 0^\circ$).

11. **Rules for finding the direction of force on a charge moving perpendicular to a magnetic field.**

(i) **Fleming's left hand rule.** Stretch the thumb and the first two fingers of the left hand mutually perpendicular to each other. If the forefinger points in the direction of the magnetic field, central finger in the direction of current, then the thumb gives the direction of force on the charged particle.

(ii) **Right hand palm rule.** Open the right hand and place it so that tips of the fingers point in the direction of the field $\vec{B}$ and thumb in the direction of velocity $\vec{v}$ of positive charge, then the palm faces towards the force $\vec{F}$.

12. **Definition of magnetic field.** The magnetic field at a point may be defined as the force acting on a unit charge moving with a unit velocity at right angles to the direction of the field.

13. **SI unit of magnetic field is tesla.** One tesla is that magnetic field in which a charge of 1 coulomb moving with a speed of 1 ms^{-1} at right angles to the field experiences a force of 1 newton.

$$1 \text{ tesla (T)} = 1 \text{ N A}^{-1} \text{ m}^{-1},$$

$$1 \text{ gauss (G)} = 10^{-4} \text{ T}.$$

14. **Lorentz force.** The total force, called Lorentz force, acting on a charge q moving with velocity $\vec{v}$ in an electric field $\vec{E}$ and magnetic field $\vec{B}$ is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

15. **Motion of charge inside an electric field.** If a potential difference V is applied between two parallel plates separated by distance d , then electric field set up between the plates is

$$E = \frac{V}{d}$$

The charge q of mass m experiences the electric force,

$$F_e = qE$$

Acceleration produced in the charge, $a = \frac{qE}{m}$

The moving charge follows a parabolic path inside the electric field.

16. **Motion of charge inside a magnetic field.**

(i) When $\vec{v} \perp \vec{B}$, the magnetic force on the charge makes it move along a circular path of radius,

$$r = \frac{mv}{qB}$$

(ii) When $\vec{v}$ makes angle θ with $\vec{B}$, the perpendicular component : $v_{\perp} = v \sin \theta$ of the initial velocity makes the charge move along a circular path of radius,

$$r = \frac{mv \sin \theta}{qB}$$

The parallel component : $v_{\parallel} = v \cos \theta$ of the initial velocity makes it move along the direction of the magnetic field. The resultant of the two components makes the charge move along a helical path of pitch,

$$h = \frac{2\pi mv \cos \theta}{qB}$$

17. **Cyclotron.** It is a device used to accelerate charged particles like protons, deuterons, α -particles, etc. to very high energies. Here charged particles move along a spiral path under the action of a perpendicular magnetic field and gain energy as they cross an alternating electric field again and again.

18. **Cyclotron frequency.** In a cyclotron, the frequency of the applied alternating electric field is equal to the frequency of revolution of the charged particle. This frequency is called *cyclotron frequency*.

It is given by, $f_c = \frac{qB}{2\pi m}$

where m is the mass and q is the charge of the positive ion. The cyclotron frequency is independent of both the velocity of the particle and the radius of its orbit.

19. **Maximum energy gained by positive ions.** If v_0 and r_0 are the maximum velocity and maximum radius of the circular path of the positive ions in a cyclotron, then

$$\frac{mv_0^2}{r_0} = qv_0 B \quad \text{or} \quad v_0 = \frac{qB r_0}{m}$$

$$\therefore \text{Maximum kinetic energy} = \frac{1}{2} mv_0^2 = \frac{q^2 B^2 r_0^2}{2m}$$

If V is the accelerating potential of the high frequency oscillator and the charged particle completes n revolutions before leaving the dees, then

$$\text{Maximum kinetic energy} = 2nqV.$$

20. **Force on a current carrying conductor in a magnetic field.** A conductor of length l carrying current I held in a magnetic field $\vec{B}$ at an angle θ with it, experiences a force given by

$$F = I l B \sin \theta$$

In vector notation, $\vec{F} = I(\vec{l} \times \vec{B})$

The direction of $\vec{F}$ is perpendicular to both $\vec{l}$ and $\vec{B}$ and is given by Fleming's left hand rule. The force is maximum when $\theta = 90^\circ$ and zero when $\theta = 0^\circ$ or 180° .

$$F_{\max} = I l B$$

21. **Force between two parallel infinitely long current carrying conductors.** The force per unit length between two long parallel conductors carrying currents I_1 and I_2 and separated by distance r is given by

$$f = \frac{\mu_0}{2\pi} \cdot \frac{I_1 I_2}{r}$$

The force is attractive when the currents are in the same direction and repulsive when the currents are in opposite directions.

22. **SI unit of current is ampere.** One ampere is that strength of current which, when flowing in two parallel infinitely long conductors of negligible cross-section placed in vacuum at a distance of 1 m from each other, produces between them a force of 2×10^{-7} newton per metre length.

23. **Force between two current elements.** The force between two parallel current elements dl_1 and dl_2 carrying currents I_1 and I_2 and separated by distance r much greater than their length is given by

$$dF = \frac{\mu_0}{4\pi} \frac{I_1 I_2}{r^2} \cdot \frac{dl_1 dl_2}{r^2}$$

24. **Relative sizes of electric and magnetic fields.** Under similar conditions, the magnetic forces are much smaller than the electric forces.

$$\frac{F_m}{F_e} = v_1 v_2 (\mu_0 \epsilon_0) = \frac{v_1 v_2}{c^2} \\ = \frac{10^{-5} \times 10^{-5}}{(3 \times 10^8)^2} \approx 10^{-27}$$

Here v_1 and v_2 are drift speeds of the electrons in the two conductors.

25. **Torque on current carrying coil in a magnetic field.** A rectangular coil of area A , carrying current I and capable of rotation about an axis perpendicular to the field $\vec{B}$ experiences a torque,

$$\tau = NIBA \sin \theta = m B \sin \theta$$

where N = number of turns in the coil, $m = NIA$ = magnetic dipole moment, θ = angle which the normal to the plane of the coil makes with the field $\vec{B}$.

In vector rotation, $\vec{\tau} = \vec{m} \times \vec{B}$

Torque is minimum when the plane of the coil is perpendicular to the magnetic field ($\theta = 0^\circ$).

Torque is maximum when the plane of the coil is parallel to the magnetic field ($\theta = 90^\circ$).

$$\tau_{\max} = NIBA.$$

26. **Moving coil galvanometer.** It is a device used to detect current in a circuit. It is based on the principle that a current carrying coil placed in a magnetic field experiences a current dependent torque, which tends to rotate the coil and produces angular deflection. It consists of a coil of wire of area A and N turns carrying current I to be measured. It is suspended in a radial magnetic field so that its plane always remains parallel to $\vec{B}$ ($\sin \theta = 1$) by a suspension fibre of torsion constant k . In equilibrium position,

Restoring torque = Deflecting torque

$$\text{or } k \alpha = NIBA \quad \text{or } \alpha = \frac{NBA}{k} \cdot I$$

i.e., Deflection of coil $\propto$ Current in the coil.

27. **Figure of merit of a galvanometer.** It is the current which produces a deflection of one scale division in the galvanometer. It is given by

$$G = \frac{I}{\alpha} = \frac{k}{NBA}$$

28. **Current sensitivity of a galvanometer.** It is the deflection produced in a galvanometer when a unit current flows through it.

$$\text{Current sensitivity} = \frac{\alpha}{I} = \frac{NBA}{k}$$

29. **Voltage sensitivity of a galvanometer.** It is the deflection produced in a galvanometer when a unit potential difference is applied across its ends.

$$\text{Voltage sensitivity} = \frac{\alpha}{V} = \frac{\alpha}{IR} = \frac{NBA}{kR}$$

30. **Conversion of a galvanometer into an ammeter.**

An ammeter is an instrument used to measure electric current in a circuit. A galvanometer of resistance G can be converted into an ammeter of range $0 - I$ by connecting a small resistance S in parallel with it. The value of small resistance, called shunt, is given by

$$S = \frac{I_g}{I - I_g} \times G$$

where I_g is the current with which galvanometer gives full-scale deflection.

Total resistance of an ammeter,

$$R_A = \frac{GS}{G + S}$$

An ammeter is a low resistance device and is connected in series in a circuit.

31. **Conversion of galvanometer into a voltmeter.** A voltmeter is used to measure potential difference between any two points of a circuit. A galvanometer of resistance G can be converted into a voltmeter of range $0 - V$ by connecting a large resistance R in series with it. The value of R is given by

$$R = \frac{V}{I_g} - G$$

Total resistance of a voltmeter,

$$R_V = G + R$$

A voltmeter is a high resistance device and is always connected parallel to the conductor across which p.d. is to be measured.

COMPETITION SECTION

Magnetism

GLIMPSES

1. **Magnets and magnetism.** A magnet is a piece of material that has both attractive and directive properties. It attracts small pieces of iron, nickel, cobalt, etc. This property of attraction is called magnetism.
2. **Natural magnets.** A natural magnet is an iron ore called lodestone (leading stone) or magnetite. It is a black iron oxide, Fe_3O_4 . Natural magnets were found as early as the sixth century B.C. in the province of Magnesia in ancient Greece, from which the word magnetism derives its name.
3. **Artificial magnets.** Pieces of iron and other magnetic materials can be made to acquire the properties of natural magnets. Such magnets are called artificial magnets.
4. **Basic properties of magnets.** These are as follows :
 - (i) *Attractive property.* A magnet attracts small pieces of iron, nickel, cobalt, etc.
 - (ii) *Directive property.* A freely suspended magnet aligns itself nearly in the geographic north-south direction.
 - (iii) *Like poles repel and unlike poles attract.* This is a fundamental law of magnetic poles.
 - (iv) *Magnetic poles exist in pairs.* Isolated magnetic poles do not exist. If we break a magnet into two pieces, we get two smaller dipole magnets.
5. **Magnetic field.** The space around a magnet within which its influence can be experienced is called its magnetic field.
6. **Uniform magnetic field.** A magnetic field in a region is said to be uniform if it has same magnitude and direction at all points of that region.
7. **Magnetic poles.** These are the regions of apparently concentrated magnetic strength in a magnet where the magnetic attraction is maximum.
8. **Magnetic axis.** The line passing through the poles of a magnet is called its magnetic axis.
9. **Magnetic equator.** The line passing through the centre of the magnet and at right angles to the magnetic axis is called the magnetic equator of the magnet.
10. **Magnetic length.** The distance between the two poles of a magnet is called its magnetic length. It is slightly less than the geometrical length of the magnet.
11. **Coulomb's law of magnetic force.** This law states that the force of attraction or repulsion between two magnetic poles is directly proportional to the product of their pole strengths and inversely proportional to the square of the distance between them. If q_{m_1} and q_{m_2} are the pole strengths of two magnetic poles separated by distance r , then the force of attraction or repulsion between them is
$$F = \frac{\mu_0}{4\pi} \cdot \frac{q_{m_1} q_{m_2}}{r^2}$$
where μ_0 is the permeability of free space and its value is $4\pi \times 10^{-7}$ henry/metre or TmA^{-1} .

12. **Magnetic dipole.** Any arrangement of two equal and opposite magnetic poles separated by a small distance is called a magnetic dipole. A bar magnet and a current-carrying loop are magnetic dipoles.

13. **Magnetic dipole moment.** It is equal to the product of the pole strength (q_m) and the magnetic length ($2l$) of the magnet.

$$m = q_m \times 2l$$

The SI unit of magnetic dipole moment is Am^2 or JT^{-1} .

14. **Magnetic lines of force.** A magnetic line of force may be defined as the curve the tangent to which at any point gives the direction of the magnetic field at that point. It may also be defined as the path along which a unit north pole would tend to move if free to do so.

15. **Properties of lines of force.** These are as follows :

- Magnetic lines of force are closed curves which start in air from the N-pole and end at the S-pole and then return to the N-pole through the interior of the magnet.
- The lines of force never cross each other.
- They start from, and end on the surface of the magnet normally.
- The lines of force have a tendency to contract lengthwise and expand sideways. This explains attraction between unlike poles and repulsion between like poles.
- The relative closeness of the lines of force is a measure of the strength of the magnetic field which is maximum at the poles.

16. **Magnetic field of a bar magnet at an axial point (end-on position).**

$$(i) B_{\text{axial}} = \frac{\mu_0}{4\pi} \cdot \frac{2mr}{(r^2 - l^2)^2},$$

where r is the distance of the point from the centre of the magnet.

- (ii) For a short magnet, $l \ll r$,

$$B_{\text{axial}} = \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3}$$

The magnetic field at any axial point of magnetic dipole is in the same direction as that of its magnetic dipole moment.

17. **Magnetic field of a bar magnet at an equatorial point (Broadside-on position).**

$$(i) B_{\text{equa}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{(r^2 + l^2)^{3/2}}$$

- (ii) For short magnet, $l \ll r$,

$$B_{\text{equa}} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}.$$

The magnetic field at any equatorial point of a magnetic dipole is in the direction opposite to that of its magnetic dipole moment.

18. **Torque on a magnet in a magnetic field.** If a magnet of dipole moment $\vec{m}$ is placed in a magnetic field $\vec{B}$ making an angle θ with it, then torque acting on the magnet is

$$\tau = mB \sin \theta$$

In vector notation, $\vec{\tau} = \vec{m} \times \vec{B}$

The effect of the torque is to align the dipole parallel to the field $\vec{B}$.

If $\theta = 90^\circ$, then $\tau = mB$

Hence the magnetic dipole may be defined as the torque acting on a magnetic dipole placed perpendicular to a uniform magnetic field of unit strength.

19. **Potential energy of a magnetic dipole in a magnetic field.** When a magnetic dipole is rotated in a magnetic field against the torque from initial position θ_1 to final position θ_2 , the work done or the potential energy stored is given by

$$W = U = -mB(\cos \theta_2 - \cos \theta_1)$$

P.E. is zero when $\vec{m} \perp \vec{B}$. Hence P.E. of the dipole in any orientation θ is

$$U = -mB \cos \theta = -\vec{m} \cdot \vec{B}$$

Special Cases

- When $\theta = 0^\circ$, $U = -mB$. Thus the P.E. of a dipole is minimum when $\vec{m}$ is parallel to $\vec{B}$. This is the position of stable equilibrium.
- When $\theta = 90^\circ$, $U = 0$.
- When $\theta = 180^\circ$, $U = +mB$.

Thus the P.E. of the dipole is maximum when $\vec{m}$ is antiparallel to $\vec{B}$. This is the position of unstable equilibrium.

20. **Current loop as a magnetic dipole.** A planar current loop of area A and carrying current I behaves as a magnetic dipole of dipole moment,

$$m = IA$$

In vector notation, $\vec{m} = I \vec{A}$

The direction of $\vec{m}$ is given by *right hand thumb rule*. If we curl the fingers of the right hand along the direction of current in the loop, then the extended thumb gives the direction of the magnetic moment associated with the loop.

21. **Magnetic dipole moment of a revolving electron.** The orbital magnetic moment of an electron revolving around a nucleus in n th orbit of radius r with speed v is given by

$$\mu_l = \frac{evr}{2} = \frac{e}{2m_e} l = n \left(\frac{eh}{4\pi m_e} \right)$$

where l is the magnitude of the angular momentum of the electron revolving around the nucleus.

22. **Bohr magneton.** It is the magnetic dipole moment associated with an electron due to its orbital motion in the first orbit of hydrogen atom. It is the smallest value of μ_l .

$$\begin{aligned} \mu_B = (\mu_l)_{\min} &= \frac{eh}{4\pi m_e} \\ &= 9.27 \times 10^{-24} \text{ Am}^2 \end{aligned}$$

23. **Gauss's law in magnetism.** This law states that the net magnetic flux through any closed surface is zero.

Mathematically,

$$\phi_B = \oint_S \vec{B} \cdot d\vec{S} = 0.$$

This law indicates that isolated magnetic poles (also called monopoles) do not exist.

24. **Basic difference between electric and magnetic lines of force.** The magnetic lines of force are continuous and form closed loops. They do not start or end at a point. In contrast, the electric lines of force start from a positive charge and end on a negative charge or they fade out at infinity.

25. **Earth's magnetic field.** When a magnet is suspended freely, it orients itself roughly in the geographical north-south direction. This suggests that the earth behaves as a huge

magnet. Its field can be approximated to that of a magnetic dipole with dipole moment $8.0 \times 10^{22} \text{ Am}^2$ with its axis aligned at a small angle with the rotation axis of the earth. The magnitude of the field on the earth's surface is typically about 10^{-4} T .

26. **Magnetic meridian.** The vertical plane passing through the magnetic axis of a freely suspended small magnet is called magnetic meridian. The earth's magnetic field acts in the direction of the magnetic meridian.
27. **Geographic meridian.** The vertical plane passing through the geographic north and south poles is called geographic meridian.
28. **Elements of earth's magnetic field.** The earth's magnetic field at a place can be completely described by three parameters which are called elements of earth's magnetic field. These are declination, dip and horizontal component of earth's magnetic field.
29. **Magnetic declination (α).** It is the angle between the geographic meridian and the magnetic meridian at the given place.
30. **Angle of dip (δ).** It is the angle made by the earth's total magnetic field with the horizontal direction.
31. **Horizontal component of earth's magnetic field.**

It is the component of the earth's total magnetic field B acting in the horizontal direction.

$$B_H = B \cos \delta$$

At the magnetic equator, $\delta = 0^\circ$,

$$B_H = B \cos 0^\circ = B$$

At the magnetic poles, $\delta = 90^\circ$,

$$B_H = B \cos 90^\circ = 0.$$

32. **Relations between the elements of earth's magnetic field.**

$$B_H = B \cos \delta$$

and $B_V = B \sin \delta$

$$\frac{B_V}{B_H} = \tan \delta$$

and $B = \sqrt{B_H^2 + B_V^2}$

33. **Neutral point.** It is the point where the magnetic field of a bar magnet is completely cancelled by

the horizontal component of earth's magnetic field.

- (i) For a magnet placed with its N-pole pointing geographic north, the neutral points lie at the equatorial line.

At each neutral point,

$$B_H = \frac{\mu_0}{4\pi} \cdot \frac{m}{(r^2 + l^2)^{3/2}}$$

$$\approx \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3} \quad \text{[For a short magnet]}$$

- (ii) For a magnet placed with its N-pole pointing geographic south, the neutral points lie at the axial line. At each neutral point,

$$B_H = \frac{\mu_0}{4\pi} \cdot \frac{2mr}{(r^2 - l^2)^2}$$

$$\approx \frac{\mu_0}{4\pi} \cdot \frac{2m}{r^3} \quad \text{[For a short magnet]}$$

34. **Magnetising field.** The magnetic field that exists in vacuum and induces magnetism is called magnetising field. The magnetising field set up in a solenoid carrying current I and placed in vacuum,

$$B_0 = \mu_0 nI$$

SI unit of B_0 is tesla (T).

35. **Magnetising field intensity or magnetic intensity.** It is the number of ampere-turns (nI) flowing round the unit length of the solenoid required to produce a given magnetising field.

Thus $H = nI$

Also, $B_0 = \mu_0 nI = \mu H$

$$\therefore H = \frac{B_0}{\mu_0}$$

SI unit of H is Am^{-1} and its dimensions are $[\text{L}^{-1}\text{A}]$.

36. **Magnetisation or intensity of magnetisation.** It is the magnetic moment developed per unit volume of a material when placed in a magnetising field. It is a vector quantity.

$$M = \frac{m}{V}$$

Its SI unit is Am^{-1} or N Wb^{-1} or $\text{Nm}^{-2} \text{T}^{-1}$.

37. **Magnetic induction.** It is the total number of magnetic lines of force crossing per unit area

through a magnetic material. Its SI unit is tesla (T).

$$B = \mu_0 (H + M)$$

38. **Magnetic permeability.** It is the ratio of the magnetic induction to the magnetising field intensity.

$$\mu = \frac{B}{H}$$

Its SI unit is Tm A^{-1} or $\text{Wbm}^{-1} \text{A}^{-1}$.

39. **Relative permeability.** It is the ratio of the permeability of the material to the permeability of free space.

$$\mu_r = \frac{\mu}{\mu_0}$$

40. **Magnetic susceptibility.** It is the ratio of the intensity of magnetisation (M) induced to the magnetising field intensity (H).

$$\chi_m = \frac{M}{H}$$

It can be shown that

$$\mu = \mu_0 (1 + \chi_m)$$

and

$$\mu_r = 1 + \chi_m$$

41. **Classification of magnetic materials.** Magnetic materials are broadly classified as diamagnetic, paramagnetic and ferromagnetic.

42. **Diamagnetic substances.** These are the substances which when placed in a magnetising field get feebly magnetised in the opposite direction of the applied field. Such substances are feebly repelled by magnets and tend to move slowly from stronger to weaker parts of a magnetic field.

Examples are Bi, Cu, Pb, Si, N_2 (at STP), H_2O and NaCl.

43. **Paramagnetic substances.** These are the substances which when placed in a magnetising field get feebly magnetised in the direction of the magnetising field. Such substances are feebly attracted by magnets and tend to move slowly from weaker to stronger parts of a magnetic field. Examples are Al, Na, Ca, O_2 (at STP) and CuCl_2 .

44. **Ferromagnetic substances.** These are the substances which when placed in a magnetising field get strongly magnetised in the direction of

the magnetising field. Such substances tend to move quickly from weaker to stronger parts of a field. Examples are Fe, Ni, Co, Gd, etc.

The magnetic susceptibility χ is small and negative for diamagnetic substances, small and positive for paramagnetic substance and large and positive for ferromagnetic substances. The relative permeability ($\mu_r = 1 + \chi$) is slightly less than 1 for diamagnetic substances, slightly greater than 1 for paramagnetic substances and of the order of thousands for ferromagnetic substances.

45. **Curie's law.** The magnetic susceptibility of a paramagnetic substance varies inversely with its absolute temperature.

$$\chi_m \propto \frac{1}{T}$$

or
$$\chi_m = \frac{C}{T},$$

where C is curie constant.

46. **Curie temperature.** The temperature above which a ferromagnetic substance becomes paramagnetic is called Curie temperature (T_C). The modified Curie law for ferromagnetic substances above the Curie temperature is

$$\chi_m = \frac{C'}{T - T_C} \quad (T > T_C)$$

This relation is also called *Curie-Weiss law*.

47. **Hysteresis.** The phenomenon of lagging of the magnetic induction behind the magnetising field in a ferromagnetic material is called hysteresis. The area of the hysteresis (B - H) loop gives the energy wasted in a sample when it is taken through a cycle of magnetisation.
48. **Retentivity or remanence.** The magnetic induction left behind in the sample after the magnetising field has been removed is called retentivity.
49. **Coercivity.** The value of the reverse magnetising field required to make the residual magnetism of a sample equal to zero is called coercivity.
50. **Tangent law.** When a short magnet is suspended freely under the combined action of two uniform perpendicular magnetic fields B

and B_H , the magnet comes to rest making an angle θ with the direction of B_H such that $B = B_H \tan \theta$.

51. **Tangent galvanometer.** Its working is based on tangent law and is used to measure very small currents. It consists of a circular coil of radius R and N turns. If a current I in the coil produces deflection θ in the compass needle at the centre of the coil, then

$$I = \left(\frac{2 R B_H}{\mu_0 N} \right) \tan \theta = K \tan \theta$$

where $K = \frac{2 R B_H}{\mu_0 N}$, is the *reduction factor* of the tangent galvanometer.

52. **Vibration magnetometer.** It is used to compare magnetic moments of two bar magnets or to determine the horizontal component of earth's magnetic field.

If a bar magnet of magnetic moment m and moment of inertia I oscillates under the horizontal component (B_H) of earth's magnetic field, then its period of vibration is

$$T = 2\pi \sqrt{\frac{I}{m B_H}}$$

53. **Uses of vibration magnetometer.**

- (i) Measurement of magnetic moment of a magnet,

$$m = \frac{4\pi^2 I}{B_H T^2}$$

- (ii) Comparison of magnetic moments of two magnets of same size and same mass.

$$\frac{m_2}{m_1} = \frac{T_1^2}{T_2^2}$$

- (iii) Comparison of magnetic moments of two magnets of unequal sizes and masses.

$$\frac{m_1}{m_2} = \frac{T_2^2 + T_1^2}{T_2^2 - T_1^2}$$

- (iv) Comparison of horizontal component of earth's magnetic field.

$$\frac{B_H}{B'_H} = \frac{T'^2}{T^2}$$

COMPETITION SECTION

Electromagnetic Induction

GLIMPSES

1. **Magnetic flux.** The number of magnetic lines of force crossing a surface normally is called magnetic flux linked with the surface. If the normal drawn to the surface area A makes angle θ with the field $\vec{B}$, then the magnetic flux is

$$\phi = BA \cos \theta = \vec{B} \cdot \vec{A}$$

Magnetic flux is a scalar quantity. Its dimensions are $[ML^2A^{-1}T^{-2}]$.

2. **SI unit of magnetic flux is weber (Wb).** It is the flux produced when a uniform magnetic field of 1 T crosses normally an area of 1 m^2 . The CGS unit of magnetic flux is *maxwell*.

$$1 \text{ weber} = 10^8 \text{ maxwell}$$

3. **Electromagnetic induction.** It is the phenomenon of production of induced emf (and hence induced current) due to a change of magnetic flux linked with a closed circuit. The term electromagnetic induction means inducing electricity by magnetism.

4. **Faraday's laws of electromagnetic induction.**
First law. Whenever the magnetic flux linked with a closed circuit changes, an emf is induced in it which lasts only so long as the change in flux is taking place.

Second law. The magnitude of induced emf is equal to the rate of change of magnetic flux linked with the closed circuit. Mathematically,

$$|\mathcal{E}| = \frac{d\phi}{dt}$$

5. **Lenz's law.** It states that the direction of induced current is such that it opposes the cause which produces it *i.e.*, it opposes the change in flux.

6. **Mathematical form of the laws of electromagnetic induction.** For a coil of one turn, induced emf is given by

$$\mathcal{E} = -\frac{d\phi}{dt} = -\frac{\phi_2 - \phi_1}{t}$$

$$\text{For a coil of } N \text{ turns, } \mathcal{E} = -N \frac{d\phi}{dt} = -N \frac{\phi_2 - \phi_1}{t}$$

The negative sign shows that induced emf opposes the change in flux. It is in accordance with Lenz's law.

7. **Motional emf.** The emf induced across the ends of a conductor due to its motion in a magnetic field is called motional emf. It is produced due to the magnetic Lorentz force acting on the free electrons of the conductor. If a conductor of length l moves with velocity v in a magnetic field B perpendicular to both its length and the direction of the magnetic field, then the emf induced across its ends is given by $\mathcal{E} = Blv$

$$\text{Induced current, } I = \frac{\mathcal{E}}{R} = \frac{Blv}{R}$$

Force necessary to move the conductor,

$$F = \frac{B^2 l^2 v}{R}$$

Power dissipated as Joule heating loss,

$$P = Fv = \frac{B^2 l^2 v^2}{R}$$

8. **Fleming's right hand rule.** It gives the direction of induced emf. If we stretch the thumb and the first two fingers of our right hand mutually perpendicular to each other and if the forefinger points in the direction of the magnetic field, thumb in the direction of motion of the conductor, then the central finger points in the direction of current induced in the conductor.

9. **Relation between induced charge and change in magnetic flux.** The induced charge flowing through a circuit depends on the net change in magnetic flux linked with circuit and is independent of the time interval of the flux change.

$$\Delta q = \frac{\Delta \phi}{R} = \frac{\text{Net change in magnetic flux}}{\text{Resistance}}$$

10. **Methods of generating induced emf.** As $\phi = BA \cos \theta$, so the magnetic flux linked with a loop can be changed and hence induced emf can be produced by three methods : (i) by changing the magnetic field B , (ii) by changing the area A of the loop and (iii) by changing the relative orientation θ of the loop and the magnetic field.

11. **Motional emf in a rotating coil.** If an N turn coil of area A is rotated with uniform angular velocity ω in a uniform magnetic field B about an axis perpendicular to the field B , then the motional emf set up across the ends of the coil is given by

$$\mathcal{E} = NBA \omega \sin \omega t = \mathcal{E}_0 \sin \omega t = \mathcal{E}_0 \sin 2\pi ft$$

where $\mathcal{E}_0 = NBA \omega$ = peak value of induced emf. Both the direction and magnitude of the induced emf change regularly with time. Such a sinusoidally varying emf is called alternating emf.

12. **Eddy currents.** These are the currents induced in solid metallic masses when the magnetic flux threading through them changes. Such currents flow in the form of irregularly shaped loops throughout the body of the metal and their direction is given by Lenz's law. Eddy currents cause unnecessary heating and wastage of power. They are reduced by using laminated soft iron cores. They are useful in (i) electric brakes, (ii) speedometers, (iii) induction furnaces and (iv) electromagnetic shielding.

13. **Self induction.** It is phenomenon of production of induced emf in a coil when a changing current passes through it.

14. **Self-inductance or coefficient of self induction (L).** When a current I flows through a coil, flux linked with it is $\phi = LI$

$$\text{Induced emf, } \mathcal{E} = -\frac{d\phi}{dt} = -L \frac{dI}{dt}$$

Thus self-inductance of a coil is the induced emf set up in it when the current passing through it changes at the unit rate. It is a measure of the opposition of the coil to the flow of current through it.

15. **Self-inductance of a long solenoid.** The self-inductance of a long solenoid of length l , area of cross-section A and having N turns is

$$L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 l A, \quad \text{where } n = \frac{N}{l}$$

When the solenoid is wound over a soft iron core of relative permeability μ_r , $L = \mu_r \mu_0 n^2 l A$

The value of self inductance depends on the number of turns in the solenoid, its area of cross-section and the relative permeability of its core material.

16. **Mutual induction.** It is the phenomenon of production of induced emf in one coil when the current through the neighbouring coil changes at the unit rate.

17. **Mutual-inductance or coefficient of mutual induction (M).** If a current I flowing through one coil generates flux ϕ in the neighbouring coil, then $\phi = MI$

$$\text{Induced emf, } \mathcal{E} = -\frac{d\phi}{dt} = -M \frac{dI}{dt}$$

Thus mutual-inductance of two coils may be defined as the induced emf set up in one coil when the current in the neighbouring coil changes at the unit rate.

18. **Mutual inductance of two long solenoids.** The mutual inductance of two long co-axial solenoids wound over one another is

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \mu_0 n_1 n_2 A l = \mu_0 n_1 n_2 \pi r_1^2 l$$

where n_1, n_2 are the number of turns per unit length of the solenoids, l is their common length and $A = \pi r_1^2$ = cross-sectional area of the inner solenoid.

19. **Henry.** It is SI unit for both self and mutual-inductances. Inductance is one henry if an induced emf of 1 volt is set up when the current changes at the rate of 1 ampere per second.

$$1 \text{ henry (H)} = 1 \text{ VsA}^{-1} = 1 \text{ Wb A}^{-1}.$$

20. **Coefficient of coupling.** It gives a measure of the manner in which two coils are coupled together. If L_1 and L_2 are the self-inductances of the two coils and M is the mutual inductance, then the coefficient of coupling

$$K = \sqrt{\frac{M}{L_1 L_2}}$$

The value of K lies between 0 and 1.

COMPETITION SECTION

Alternating Current and Electrical Machines

GLIMPSES

1. **Alternating current.** It is that current which varies in magnitude continuously and reverses its direction periodically. Its value at any instant is given by

$$I = I_0 \sin \omega t = I_0 \sin 2\pi ft$$

where I_0 is the peak value of a.c. or current amplitude. The frequency of a.c. supplied to our homes is 50 cps. The average value of a.c. over a complete cycle is zero.

2. **Direct current.** It is that current which flows with a constant magnitude in the same fixed direction.
3. **Average or mean value of a.c.** It is that value of direct current which sends the same charge in a circuit in the same time as is sent by the given alternating current in the same circuit in its half time period.

$$I_{av} = \frac{2}{\pi} I_0 = 0.637 I_0$$

4. **Effective or rms or virtual value of a.c.** It is that value of direct current which produces the same heating effect in a given resistor as is produced by the given alternating current when passed for the same time.

$$I_{rms} \text{ or } I_{eff} \text{ or } I_v = \frac{1}{\sqrt{2}} \cdot I_0 = 0.707 I_0$$

5. **Alternating voltage.** It is that voltage whose magnitude varies continuously and direction reverses periodically with time. Its instantaneous, average and root mean square values are respectively given by

$$\mathcal{E} = \mathcal{E}_0 \sin \omega t, \quad \mathcal{E}_{av} = 0.637 \mathcal{E}_0,$$

$$\mathcal{E}_{rms} = \frac{1}{\sqrt{2}} \mathcal{E}_0$$

6. **Phasors and phasor diagrams.** A rotating vector that represents a sinusoidally varying quantity is called a phasor. A diagram that represents alternating current and voltage of the same frequency as rotating vectors (phasors) along with proper phase angle between them is called a phasor diagram or Argand diagram.

7. **A.C. circuit containing resistor only.** An alternating voltage, $\mathcal{E} = \mathcal{E}_0 \sin \omega t$ applied to a resistor R drives a current $I = I_0 \sin \omega t$ in the resistor. The current is in phase with the applied voltage.

$$\text{Peak value of current, } I_0 = \frac{\mathcal{E}_0}{R}$$

8. **A.C. circuit containing only an inductor.** An alternating voltage, $\mathcal{E} = \mathcal{E}_0 \sin \omega t$ applied to a pure inductor L drives a current, $I = I_0 \sin (\omega t - \pi/2)$ in the inductor. The current in the inductor lags behind the voltage in phase by $\pi/2$ rad.

$$\text{Peak value of current, } I_0 = \frac{\mathcal{E}_0}{\omega L} = \frac{\mathcal{E}_0}{X_L}$$

Root mean square value of current,

$$I_{rms} = \frac{\mathcal{E}_{rms}}{X_L} = \frac{\mathcal{E}_{rms}}{\omega L} = \frac{\mathcal{E}_0}{\sqrt{2} \cdot \omega L}$$

9. **Reactance.** The non-resistive opposition to the flow of a.c. is called reactance. It may be inductive reactance (X_L) or capacitive reactance (X_C).
10. **Inductive reactance.** It is the effective resistance or opposition offered by an inductor to the flow of a.c. through it. It is given by

$$X_L = \omega L = 2\pi fL$$

The SI unit of inductive reactance is ohm (Ω).

For a.c., $X_L \propto f$

For d.c., $f = 0$, so $X_L = 0$.

11. **A.C. circuit containing only a capacitor.** An alternating voltage, $\mathcal{E} = \mathcal{E}_0 \sin \omega t$ applied to a capacitor C drives a current, $I = I_0 \sin (\omega t + \pi/2)$ in the capacitor. The current in the inductor is ahead of voltage in phase by $\pi/2$ rad.

Peak value of current,

$$I_0 = \frac{\mathcal{E}_0}{1/\omega C} = \frac{\mathcal{E}_0}{X_C}$$

Root mean square value of current,

$$I_{rms} = \frac{\mathcal{E}_{rms}}{X_C} = \frac{\mathcal{E}_{rms}}{1/\omega C} = \frac{\mathcal{E}_0}{\sqrt{2} \cdot 1/\omega C}$$

12. **Capacitive reactance.** It is the effective resistance or opposition offered by a capacitor to the flow of a.c. through it.

It is given by

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

The SI unit of capacitive reactance is ohm (Ω)

For a.c., $X_C \propto \frac{1}{f}$

For d.c., $f = 0$, so $X_C = \infty$

Thus a capacitor allows a.c. to flow through it easily but blocks d.c.

13. **Impedance.** It is a quantity that measures the opposition of a circuit to the flow of current through it and so determines the magnitude of the current. In a d.c. circuit, this is the resistance R alone. In an a.c. circuit, the reactance X also has to be taken into account, according to the relation :

$$Z^2 = R^2 + X^2$$

where Z is the impedance. **Impedance triangle** is a right angled triangle whose base represents resistance R , perpendicular represents reactance X and hypotenuse represents impedance Z of the circuit. From this triangle, the phase angle ϕ between voltage and current is given by

$$\tan \phi = \frac{X}{R} \quad \text{or} \quad \cos \phi = \frac{R}{Z}$$

14. **A.C. through a series LR-circuit.** The alternating voltage leads the current by a phase angle ϕ

Instantaneous voltage, $\mathcal{E} = \mathcal{E}_0 \sin \omega t$

Instantaneous current, $I = I_0 \sin (\omega t - \phi)$

Peak current,

$$I_0 = \frac{\mathcal{E}_0}{\sqrt{R^2 + X_L^2}} = \frac{\mathcal{E}_0}{Z}$$

Impedance,

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + \omega^2 L^2}$$

Phase angle,

$$\phi = \tan^{-1} \frac{X_L}{R} \quad \text{or} \quad \phi = \cos^{-1} \frac{R}{Z}$$

15. **A.C. through a series CR-circuit.** The alternating voltage lags behind the voltage by phase angle ϕ

Instantaneous voltage, $\mathcal{E} = \mathcal{E}_0 \sin \omega t$

Instantaneous current, $I = I_0 \sin (\omega t + \phi)$

Peak current, $I_0 = \frac{\mathcal{E}_0}{\sqrt{R^2 + X_C^2}} = \frac{\mathcal{E}_0}{Z}$

Impedance, $Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + \omega^2 L^2}$

Phase angle, $\phi = \tan^{-1} \frac{X_C}{R} \quad \text{or} \quad \phi = \cos^{-1} \frac{R}{Z}$

16. **The series LCR-circuit.** For a series LCR-circuit connected across a source $\mathcal{E} = \mathcal{E}_0 \sin \omega t$, the current I is given by

$$I = I_0 \sin (\omega t - \phi) = \frac{\mathcal{E}_0}{Z} \sin (\omega t - \phi)$$

where Z is the total effective resistance of the LCR-circuit and is called its impedance.

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

The phase angle ϕ between voltage and current is given by

$$\tan \phi = \frac{X_L - X_C}{R} \quad \text{and} \quad \cos \phi = \frac{R}{Z}$$

The voltage leads the current if $X_L > X_C$ and it lags behind the current if $X_L < X_C$.

17. **Resonance condition of the LCR-circuit.** If $X_L = X_C$, the impedance of LCR circuit becomes

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = R$$

The impedance is minimum and hence current is maximum. The circuit is purely resistive and voltage and current are in same phase. This is the resonance condition of LCR-circuit and is satisfied at the resonant frequency given by

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

18. **Q-Factor.** It indicates the sharpness of resonance. It is defined as the ratio of the resonant frequency to the difference in two frequencies taken on both sides of the resonant frequency such that at each frequency, the current amplitude becomes $1/\sqrt{2}$ times the value at resonant frequency.

$$\begin{aligned} \text{Q-factor} &= \frac{\text{Resonant frequency}}{\text{Band width}} \\ &= \frac{\omega_r}{\omega_2 - \omega_1} = \frac{\omega_r L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} \end{aligned}$$

19. **Power in an a.c.-circuit.** The average power in an a.c. circuit consumed per cycle is given by

$$P_{av} = \mathcal{E}_{rms} I_{rms} \cos \phi = \frac{\mathcal{E}_0 I_0}{2} \cos \phi$$

Here $\cos \phi$ is called **power factor** which is the ratio of true power (P_{av}) to the apparent power ($\mathcal{E}_{rms} I_{rms}$).

$$\text{Power factor, } \cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

- (i) For a pure resistive circuit, $\phi = 0$,

$$P_{av} = \mathcal{E}_{rms} I_{rms} = \frac{\mathcal{E}_{rms}^2}{R}$$

- (ii) For a pure inductive circuit, $\phi = \frac{\pi}{2}$, $P_{av} = 0$.

- (iii) For a pure capacitive circuit, $\phi = -\frac{\pi}{2}$, $P_{av} = 0$.

20. **Wattless current.** The current in an a.c. circuit is said to be wattless if the average power consumed in the circuit is zero. It is the component $I_{rms} \sin \phi$ of the alternating current. In an inductive or capacitive a.c. circuit, the phase difference between voltage and current, $\phi = \pm \pi/2$. Power factor $\cos \phi = 0$ and so the current is wattless.
21. **Energy stored in an inductor.** When the current in an inductor grows from 0 to I_0 , the magnetic energy stored in it is

$$U = \frac{1}{2} L I_0^2$$

22. **Energy stored in a capacitor.** The energy stored in a capacitor when it is charged from 0 to V volt is

$$U = \frac{1}{2} C V^2 = \frac{1}{2} Q V = \frac{1}{2} \frac{Q^2}{C}$$

23. **LC-oscillations.** When a charged capacitor is allowed to discharge through a non-resistive

inductor, electrical oscillations of constant amplitude and frequency are produced. These oscillations are called LC-oscillations. The charge of the capacitor satisfies the equation of SHM :

$$\frac{d^2 q}{dt^2} + \frac{1}{LC} q = 0$$

Instantaneous charge, $q = q_0 \cos \omega_0 t$

Instantaneous current, $I = -\frac{dq}{dt} = \omega_0 q_0 \sin \omega_0 t$

Angular frequency of free oscillations,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Frequency of free oscillations,

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

The energy in the LC-circuit oscillates between the capacitor and the inductor but the total energy remains constant.

24. **Choke coil.** It is an inductor with large inductance used to reduce current in an a.c. circuit without much loss of power.

25. **Transformer.** It is a device used to convert a.c. at high voltage into that at low voltage or vice versa. For an ideal transformer,

$$\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{I_1}{I_2} = \frac{N_2}{N_1} = k$$

Here suffix 1 refers to primary coil and suffix 2 to secondary coil and k is called **transformation or turns ratios** of the transformer.

- (i) For a step up transformer, $k > 1$ or $N_2 > N_1$

$$\therefore \mathcal{E}_2 > \mathcal{E}_1 \text{ and } I_2 < I_1$$

- (ii) For a step down transformer, $k < 1$ or $N_2 < N_1$

$$\therefore \mathcal{E}_2 < \mathcal{E}_1 \text{ and } I_2 > I_1$$

26. **A.C. generator.** It is a device to convert mechanical energy into electrical energy of alternating form. It consists of a coil of wire that rotates with angular velocity ω in the magnetic field B of a permanent magnet. The flux through the coil varies as $\phi = NBA \cos \omega t$, where N is the number of turns in the coil having the face area A . (At $t = 0$, the loop is normal to the field).

Induced emf, $\mathcal{E} = NBA \omega \sin \omega t = \mathcal{E}_0 \sin \omega t$

Current, $I = \frac{NBA \omega}{R} \sin \omega t = I_0 \sin \omega t$

COMPETITION SECTION

Electromagnetic Waves

GLIMPSES

1. **Displacement current.** It is that current which comes into existence (in addition of conduction current) whenever the electric field and hence the electric flux changes with time. It is equal to ϵ_0 times the rate of change of electric flux through a given surface.

$$I_D = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 A \frac{dE}{dt}$$

2. **Modified Ampere circuital law.** It states that the line integral of the magnetic field $\vec{B}$ over a closed path is equal to μ_0 times the sum of conduction current I_C and the displacement current (I_D) threading the closed path.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_C + I_D)$$
$$= \mu_0 \left(I_C + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

The sum of conduction and displacement currents has an important property of continuity i.e., the sum remains constant along any closed path.

3. **Maxwell's equations.** These are as follows :

(i) Gauss law of electrostatics :

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

(ii) Gauss law of magnetism :

$$\oint \vec{B} \cdot d\vec{S} = 0$$

(iii) Faraday's law of electromagnetic induction :

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\phi_B}{dt} = -\frac{d}{dt} \left[\oint \vec{B} \cdot d\vec{S} \right]$$

(iv) Modified Ampere's circuital law :

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[I_C + \epsilon_0 \frac{d\phi_E}{dt} \right]$$

4. **Electromagnetic wave.** An electromagnetic wave is a wave radiated by an accelerated charge and which propagates through space as coupled electric and magnetic fields, oscillating perpendicular to each other and to the direction of propagation of the wave.
5. **Principle of production of electromagnetic waves.** An accelerated charge produces electric and magnetic fields, which vary both in space and time. The two oscillating fields act as sources of each other and hence sustain each other. This results in the propagation of an electromagnetic wave through space.
6. **Source of an electromagnetic wave.** An accelerating charge produces electromagnetic waves. An electric charge oscillating harmonically with frequency ν , generates electromagnetic waves of the same frequency ν . An electric dipole is a basic source of electromagnetic waves.

An LC-circuit containing inductance L and capacitance C produces electromagnetic waves of frequency,

$$\nu = \frac{1}{2\pi\sqrt{LC}}$$

7. **Mathematical representation of electromagnetic waves.** For a plane electromagnetic wave of frequency ν , wavelength λ , propagating along

x -axis, the electric and magnetic fields may be represented as follows :

$$\vec{E} = E_y \hat{j} = E_0 \sin(kx - \omega t) \hat{j}$$

$$= E_0 \sin \left[2\pi \left(\frac{x}{\lambda} - \nu t \right) \right] \hat{j}$$

$$= E_0 \sin \left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \right] \hat{j}$$

$$E_x = E_z = 0$$

and $\vec{B} = B_z \hat{k} = B_0 \sin(kx - \omega t) \hat{k}$

$$= B_0 \sin \left[2\pi \left(\frac{x}{\lambda} - \nu t \right) \right] \hat{k}$$

$$= B_0 \sin \left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \right] \hat{k}$$

$$B_x = B_y = 0$$

Here E_0 and B_0 are the amplitudes of the electric and magnetic fields respectively.

Propagation constant,

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{c}$$

8. **Hertz's experiment.** In 1888, Hertz succeeded in experimentally confirming the existence of electromagnetic waves. By using oscillatory LC-circuits, he not only produced and detected electromagnetic waves, but also demonstrated their properties of reflection, refraction, interference and polarisation and established beyond doubt that light radiation has wave nature. He was able to produce e.m. waves of wavelength around 6 m.

9. **Contributions of J.C. Bose and Marconi.** In 1895, Sir J.C. Bose succeeded in producing and observing electromagnetic waves of much shorter wavelength (25 mm to 5 mm).

In 1896, Guglielmo Marconi succeeded in transmitting electromagnetic waves over distances of many kilometres. His experiments marked the beginning of radio communication.

10. **Basic properties of electromagnetic waves.** These are

- (i) The e.m. waves are produced by accelerated charges and do not require any medium for their propagation.

- (ii) The oscillations of $\vec{E}$ and $\vec{B}$ fields are perpendicular to each other as well as to the direction of propagation of the wave. So the e.m. waves are transverse in nature.

- (iii) The oscillations of $\vec{E}$ and $\vec{B}$ fields are in same phase.

- (iv) All electromagnetic waves travel in free space with same speed c given by

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ ms}^{-1}$$

In a material medium of refractive index n , the speed of an electromagnetic wave is given by

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{c}{\sqrt{\mu_r \epsilon_r}} = \frac{c}{n}$$

- (v) The ratio of the amplitudes of electric and magnetic fields is

$$\frac{E_0}{B_0} = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

- (vi) The electromagnetic waves carry energy as they travel through space and this energy is shared equally by the electric and magnetic fields. The average energy of an electromagnetic wave is

$$u = u_E + u_B = \frac{1}{2} \left[\epsilon_0 E_0^2 + \frac{B_0^2}{\mu} \right]$$

11. **Momentum of an electromagnetic wave.** If an electromagnetic wave transfers a total energy U to a surface in time t , then total linear momentum delivered to the surface is $p = \frac{U}{c}$

12. **Intensity of an electromagnetic wave.** The energy crossing per unit area per unit time in a direction perpendicular to the direction of propagation of the wave is called intensity of the wave.

$$\text{Intensity} = \frac{\text{Energy / time}}{\text{Area}} = \frac{\text{Power}}{\text{Area}}$$

For an e.m. wave of average energy density u ,

$$I = uc$$

$$\text{Also, } I = \frac{1}{2} \epsilon_0 E_0^2 c = \epsilon_0 E_{\text{rms}}^2 c$$

$$\text{and } I = \frac{1}{2\mu_0} B_0^2 c = \frac{1}{\mu_0} B_{\text{rms}}^2 c$$

13. **Electromagnetic spectrum.** The orderly distribution of the electromagnetic waves in accordance with their wavelength or frequency into distinct groups having widely differing properties is called electromagnetic spectrum. The main parts of an electromagnetic spectrum in the order of increasing wavelength from

$10^{-2} \text{ \AA}$ or 10^{-12} m to 10^{-6} m are γ -rays, X-rays, ultraviolet rays, visible light, infrared rays, microwaves and radiowaves. The different parts of the e.m. spectrum differ in their methods of production and detection.

COMPETITION SECTION

Ray Optics and Optical Instruments

GLIMPSES

1. **Optics.** It is the branch of physics that deals with the study of nature, production and propagation of light. It has *two* sub-branches : ray optics and wave optics.
2. **Ray or geometrical optics.** It concerns itself with the particle nature of light and is based on (i) the rectilinear propagation of light and (ii) the laws of reflection and refraction of light.
3. **Wave or physical optics.** It concerns itself with the wave nature of light and is based on the phenomena like (i) interference (ii) diffraction and (iii) polarisation of light.
4. **Laws of reflection of light.** (i) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.
(ii) The angle of incidence ' i ' is equal to the angle of reflection ' r ' i.e., $\angle i = \angle r$.
5. **Properties of images formed by plane mirrors.**
 - (i) The image formed by a plane mirror is virtual, erect and laterally reversed.
 - (ii) The size of the image is equal to the size of the object.
 - (iii) The image is as far behind the mirror as the object is in front of it.
 - (iv) The line joining the object and the image is normal to the plane mirror.
 - (v) When a plane mirror is rotated through a certain angle, the reflected ray turns through twice this angle.
6. **Images formed by inclined mirrors.** When two planes mirrors are kept facing each other at an angle θ and an object is placed between them, a

number of images are formed due to multiple reflections.

If θ is a submultiple of 180° , then the number of images formed is $n = \frac{360}{\theta} - 1$.

If θ is not a submultiple of 180° , then the number of images formed is the integer next higher than $\left(\frac{360}{\theta} - 1 \right)$. For two parallel plane mirrors,
$$n = \frac{360}{0} = \infty.$$

7. **Spherical mirror.** It is a mirror whose reflecting surface forms part of a hollow sphere. Spherical mirrors are of *two* types :
 - (i) **Concave mirror** in which the reflection of light takes place from the inner hollow surface.
 - (ii) **Convex mirror** in which the reflection of light takes place from the outer bulged surface.
8. **Definitions in connection with spherical mirrors.**
 - (i) **Pole.** It is the middle point P of the spherical mirror.
 - (ii) **Centre of curvature.** It is the centre C of the sphere of which the mirror forms a part.
 - (iii) **Radius of curvature.** It is radius (R) of the sphere of which the mirror forms a part.
 - (iv) **Principal axis.** The line PC passing through the pole and the centre of curvature of the mirror is called its principal axis.
 - (v) **Linear aperture.** It is the diameter of the circular boundary of the spherical mirror.

- (vi) **Angular aperture.** It is the angle subtended by the boundary of the spherical mirror at its centre of curvature C .
- (vii) **Principal focus.** A narrow beam of light parallel to the principal axis either actually converges to or appears to diverge from a point F on the principal axis after reflection from the spherical mirror. This point is called the principal focus of the mirror. A concave mirror has a real focus while a convex mirror has a virtual focus.
- (viii) **Focal length.** It is the distance ($f = PF$) between the focus and the pole of the mirror.
- (ix) **Focal plane.** The vertical plane passing through the principal focus and perpendicular to the principal axis is called focal plane. When a parallel beam of light is incident on a concave mirror at a small angle to the principal axis, it is converged to a point in the focal plane of the mirror.
9. **New cartesian sign convention for spherical mirrors.**
- All ray diagrams are drawn with the incident light travelling from left to right.
 - All distances are measured from the pole of the mirror.
 - All distances measured in the direction of incident light are taken positive.
 - All distances measured in the opposite direction of incident light are taken to be negative.
 - Heights measured upwards and perpendicular to the principal axis are taken positive.
 - Heights measured downwards and perpendicular to the principal axis are taken as negative.

10. **Relation between focal length and radius of curvature of a spherical mirror.**

$$\text{Focal length} = \frac{1}{2} \times \text{Radius of curvature}$$

$$\text{or} \quad f = \frac{R}{2}$$

In new cartesian sign convention, the focal length and radius of curvature are taken negative for a concave mirror and positive for a convex mirror.

11. **Spherical mirror formula.** This gives relation between object distance u , image distance v and the focal length f a spherical mirror.

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

12. **Linear or transverse magnification.** It is the ratio of the height of the image to that of the object.

$$m = \frac{\text{Height of image}}{\text{Height of object}} = \frac{h_2}{h_1} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

- If $|m| > 1$, the image is *magnified*.
 - If $|m| < 1$, the image is *diminished*.
 - If $|m| = 1$, the image is of the same size as the object.
 - If m is positive, the image is *virtual* and *erect*.
 - If m is negative, the image is *real* and *inverted*.
13. **Refraction of light.** It is the phenomenon of bending of light from its straight path when it passes at an angle from one transparent medium to another.

14. **Laws of refraction of light :**

First law. The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane.

Second law. The ratio of the sine of the angle of incidence and the sine of the angle of refraction is constant for a given pair of media. This law is also known as *Snell's law of refraction*

$$\frac{\sin i}{\sin r} = \mu, \text{ a constant.}$$

The constant μ is called refractive index of second medium w.r.t. first medium.

15. **Refractive index.** Refractive index of a medium for a light of given wavelength may be defined as the ratio of the speed of light in vacuum to its speed in that medium.

$$\mu = \frac{\text{Velocity of light in vacuum}}{\text{Velocity of light in medium}} = \frac{c}{v}$$

It may also be defined as the ratio of the wavelength of light in vacuum to its wavelength in that medium.

$$\mu = \frac{\lambda_{\text{vac}}}{\lambda_{\text{med}}}$$

The refractive index of a medium with respect to vacuum is also called **absolute refractive index**.

16. **Relative refractive index.** The relative refractive index of medium 2 w.r.t. medium 1 is the ratio of speed of light (v_1) in medium 1 to the speed of light (v_2) in medium 2.

$${}_1\mu_2 = \frac{v_1}{v_2}$$

Also
$${}_1\mu_2 = \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \text{constant}$$

or
$$\mu_1 \sin i = \mu_2 \sin r$$

17. **Principle of reversibility of light.** This principle states that if the final path of ray of light after it has suffered several reflections and refractions is reversed, it retraces its path exactly. It follows from this principle that

$${}_1\mu_2 = \frac{1}{{}_2\mu_1}$$

i.e., the refractive index of medium 2 w.r.t. medium 1 is reciprocal of the refractive index of medium 1 w.r.t. medium 2.

18. **Refraction through a rectangular glass slab.** A ray of light on refraction through a glass slab does not suffer any deviation, i.e., the incident and emergent rays are parallel, but the emergent ray is laterally displaced w.r.t. the incident ray. The lateral displacement x on passing through a glass slab of thickness t and refractive index μ is given by

$$x = \frac{t}{\cos r} \sin(i - r) = t \sin i \left[1 - \frac{\cos i}{(\mu^2 - \sin^2 i)^{1/2}} \right]$$

where i is angle of incidence

$$x_{\max} = t \sin 90^\circ = t$$

Thus the displacement of the emergent ray cannot exceed the thickness of the glass slab.

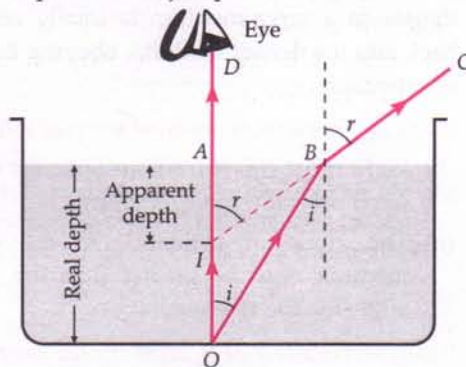
19. **Refraction through a combination of media.** When a ray of light passes through a combination of media, the quantity $\mu \sin i$ remains constant, where μ is the absolute refractive index of the medium and i the angle of incidence in that medium. Thus

$$\begin{aligned} \mu_{\text{air}} \times \sin i_{\text{air}} &= \mu_{\text{glass}} \times \sin i_{\text{glass}} \\ &= \mu_{\text{water}} \times \sin i_{\text{water}} \end{aligned}$$

Also ${}^a\mu_\omega \times {}^\omega\mu_g \times {}^g\mu_a = 1$

and
$${}^\omega\mu_g = \frac{{}^a\mu_g}{{}^a\mu_\omega}$$

20. **Relation between real depth and apparent depth.** Due to refraction of light, the apparent depth of an object placed in a denser medium is



less than the real depth. When an object O , in a denser medium of thickness t and refractive index μ is seen through a rarer medium, its image is seen at I . It is seen that

$$\mu = \frac{\text{Real depth}}{\text{Apparent depth}} = \frac{AO}{AI}$$

Also, apparent depth, $AI = \frac{t}{\mu}$

The height through which an object appears to be raised in a denser medium is called **normal shift**.

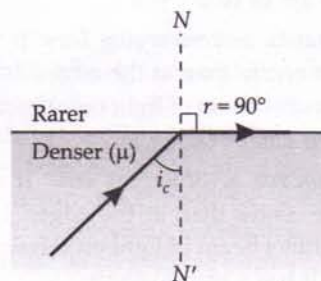
$$\therefore \text{Normal shift, } d = IO = AO - AI = t \left(1 - \frac{1}{\mu} \right)$$

Total normal shift for compound media

$$= t_1 \left(1 - \frac{1}{\mu_1} \right) + t_2 \left(1 - \frac{1}{\mu_2} \right) + \dots$$

21. **Critical angle and total internal reflection.** The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90° is called **critical angle** of the denser medium and is denoted by i_c . When $i = i_c$, $r = 90^\circ$.

As
$$\frac{\sin i_c}{\sin 90^\circ} = \frac{1}{\mu} \quad \text{or} \quad \mu = \frac{1}{\sin i_c}$$



Total internal reflection is the phenomenon in which a ray of light travelling at an angle of incidence greater than the critical angle from a denser to a rarer medium is totally reflected back into the denser medium, obeying the laws of reflection.

22. Necessary conditions for total internal reflection.

- (i) Light must travel from an optically denser to an optically rarer medium.
- (ii) The angle of incidence in the denser medium must be greater than the critical angle for the two media.

23. Relation between critical angle and refractive index.
$$\mu = \frac{1}{\sin i_c}$$

24. Totally reflecting prisms. A right angled isosceles prism, i.e., a $45^\circ - 90^\circ - 45^\circ$ prism is called a totally reflecting prism. It can be used to deviate rays through 90° or 180° .

25. Mirage. It is an optical illusion observed in deserts or over hot extended surfaces like a coaltarred road due to which a traveller sees a shimmering pond of water some distance ahead him and in which the surrounding objects like tree, etc. appear inverted.

26. Optical fibres. Optical fibres consist of thousands of fine strands of quality glass, coated with a material of lower refractive index. Light entering the fibres at one end undergoes several total internal reflections and finally emerges out without any appreciable change in intensity. A bundle of optical fibres is called a light pipe, used in medical and optical examination and in receiving and transmitting signals in telecommunication.

27. Lens. A lens is a piece of a refracting medium bounded by two surfaces, at least one of which is a curved surface.

Lenses are of two types :

- (i) **Convex or converging lens.** It is thicker at the centre than at the edges. It converges a parallel beam of light on refraction through it. It has a real focus.
- (ii) **Concave or diverging lens.** It is thinner at the centre than at the edges. It diverges a parallel beam of light on refraction through it. It has a virtual focus.

28. Definitions in connection with spherical lenses :

- (i) **Centre of curvature.** The centre of curvature of the surface of a lens is centre of the sphere of which it forms a part. Because a lens has two surfaces, so it has two centres of curvature.
- (ii) **Radius of curvature.** The radius of the surface of a lens is the radius of the sphere of which the surface forms a part.
- (iii) **Principal axis.** It is the line passing through the two centres of curvature of the lens.
- (iv) **Principal focus.** A narrow beam of light parallel to the principal axis either converges to a point or appears to diverge from a point on the principal axis after refraction through the lens. This point is called principal focus. A lens has two principal foci.
- (v) **Optical Centre.** It is the point situated within the lens through which a ray of light passes undeviated.
- (vi) **Focal length.** It is the distance between the principal focus and the optical centre of the lens.
- (vii) **Aperture.** It is the diameter of the circular boundary of the lens.

29. New Cartesian sign convention for spherical lenses :

- (i) All distances are measured from the optical centre of the lens.
- (ii) The distances measured in the direction of incident light are taken as positive.
- (iii) The distances measured in the opposite direction of incident light are taken as negative.
- (iv) Heights measured upwards and perpendicular to the principal axis are taken as positive.
- (v) Heights measured downwards and perpendicular to the principal axis are taken as negative.

In this sign convention, the focal length of a converging lens is positive and that of a diverging lens is negative.

30. Refraction through a spherical surface. A surface which forms part of a sphere of a transparent refracting material is called a spherical refracting surface.

- (i) **Refraction from rarer to denser medium.** When a ray of light travels from a rarer medium of

refractive index μ_1 to a denser medium of refractive index μ_2 of a spherical surface of radius of curvature R , the relation between object distance u and image distance v is

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

If the rarer medium is air, then $\mu_1 = 1$ and $\mu_2 = \mu$, we have $\frac{\mu}{v} - \frac{1}{u} = \frac{\mu - 1}{R}$

(ii) *Refraction from denser to rarer medium.* When the object is placed in a denser medium, the relation between u and v can be obtained by interchanging μ_1 and μ_2 .

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_1 - \mu_2}{R}$$

31. **Power of a spherical refracting surface.** It is given by $P = \frac{\mu_2 - \mu_1}{R} = \frac{\mu - 1}{R}$ (for air)

where R is measured in metre. The power of a convex surface is positive and that of a concave surface is negative.

32. **Principal focal lengths of a spherical surface.**

(i) *First principal focal length.* It is the distance of a point from the pole of the surface at which if an object is placed, the image is formed at infinity.

First principal focal length, $f_1 = -\frac{\mu_1 R}{\mu_2 - \mu_1}$

(ii) *Second principal focal length.* It is the distance of a point from the pole of the surface at which the image of an object at infinity is formed.

Second principal focal length, $f_2 = \frac{\mu_2 R}{\mu_2 - \mu_1}$

33. **Lens maker's formula.** This formula relates the focal length f to the refractive index μ and the radii of curvature R_1, R_2 of its spherical surfaces.

$$\frac{1}{f} = \left[\frac{\mu_2 - \mu_1}{\mu_1} \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

For the lens placed in air,

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

34. **Thin lens formula.** This formula gives relationship between object distance u , image distance v and focal length f a spherical lens (convex or concave) of small aperture.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

35. **Linear magnification produced by a lens.** It is the ratio of the size of the image formed by a lens to the size of the object.

$$\text{Magnification} = \frac{\text{size of image}}{\text{size of object}}$$

$$\text{or } m = \frac{h_2}{h_1} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

When m is positive (or v is negative), the image is virtual and erect. When m is negative (or v is positive), the image is real and inverted.

36. **Power of a lens.** The power of a lens is defined as the reciprocal of its focal length, expressed in metres.

$$P = \frac{1}{f \text{ (m)}}$$

SI unit of power is m^{-1} , also called dioptre (D). One dioptre is the power of a lens whose principal focal length is 1 metre.

$$P = \frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

37. **Lens combinations.** When lenses are used in combination, each lens magnifies the image formed by the preceding lens. The total magnification is equal to the product of the magnifications produced by the individual lenses.

$$m = m_1 \times m_2 \times m_3 \times \dots$$

The combined focal length f of two thin lenses of focal lengths f_1 and f_2 placed in contact is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

For n thin lenses in contact,

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \dots + \frac{1}{f_n} \text{ or } P = P_1 + P_2 + \dots + P_n$$

When the two thin lenses are separated by a distance d , their equivalent focal length f is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{d}{f_1 f_2}$$

or Power, $P = P_1 + P_2 + d \times P_1 \times P_2$

38. **Prism.** A prism is a portion of a refracting medium bounded by two plane faces inclined to each other at a certain angle. The two plane faces inclined to each other are called *refracting faces*. The line along which the two refracting faces meet is called *refracting edge of the prism*. The third face of the prism opposite to the

refracting edge is called *base of the prism*. The angle included between the two refracting faces is called *angle of prism*.

39. **Refraction through a prism.** When a ray of light is refracted through a prism, the sum of the angle of incidence i and the angle of emergence i' is equal to the sum of the angle of the prism A and the angle of deviation δ .

$$A + \delta = i + i' \quad \text{and} \quad A = r + r'$$

where r and r' are the corresponding angles of refraction at the two faces.

40. **Relation between the refractive index and angle of minimum deviation.** The minimum value of the angle of deviation suffered by a ray on passing through a prism is called the angle of minimum deviation and is denoted by δ_m . When a ray of light suffers minimum deviation.

$$i = i', \quad r = r' \quad \text{and} \quad \delta = \delta_m$$

$$\therefore A + \delta_m = i + i = 2i \quad \text{or} \quad i = \frac{A + \delta_m}{2}$$

$$\text{and} \quad A = r + r = 2r \quad \text{or} \quad r = \frac{A}{2}$$

$$\text{Refractive index, } \mu = \frac{\sin i}{\sin r} = \frac{\sin \frac{A + \delta_m}{2}}{\sin \frac{A}{2}}$$

41. **Deviation produced by a prism of small angle.** It does not depend on the angle of incidence and is given by $\delta = (\mu - 1) A$.

42. **Dispersion.** The splitting of white light into its constituent colours when it passes through a glass prism is called dispersion. The dispersion of light occurs because refractive index of prism material is different for different wavelengths.

43. **Angular dispersion.** The angular separation between the two extreme colours (violet and red) in the spectrum is called angular dispersion.

Angular dispersion

$$= \delta_V - \delta_R = (\mu_V - 1) A - (\mu_R - 1) A = (\mu_V - \mu_R) A$$

44. **Dispersion power.** It is the ability of the prism material to cause dispersion and is defined as the ratio of the angular dispersion to the mean deviation.

$$\text{Dispersion power} = \frac{\text{Angular dispersion}}{\text{Mean deviation}}$$

$$\omega = \frac{\delta_V - \delta_R}{\delta} = \frac{(\mu_V - 1) A - (\mu_R - 1) A}{(\mu - 1) A} = \frac{\mu_V - \mu_R}{\mu - 1}$$

$$\text{Here} \quad \delta = \frac{\delta_V + \delta_R}{2} \quad \text{and} \quad \mu = \frac{\mu_V + \mu_R}{2}$$

45. **Pure and impure spectra.** The spectrum in which the component colours of the spectra of different rays overlap each other and the various colours are not distinctly seen is called an impure spectrum. A spectrum in which there is no overlapping of colours and different colours are distinctly seen is called the pure spectrum.

46. **Spectroscope or spectrometer.** It is an optical device used for producing and studying the spectrum of various light sources. It consists of three main parts : (i) collimator, (ii) prism table and (iii) telescope.

47. **Spherical aberration.** The inability of a lens or spherical mirror of large aperture to bring the paraxial and marginal rays of a wide beam of light to focus at a single point is called spherical aberration.

48. **Chromatic aberration.** The inability of a lens to bring the light rays of different colours to focus at a single point is called chromatic aberration.

Longitudinal chromatic aberration of a lens

$$= \text{Dispersive power}$$

$$\times \text{focal length of the lens for mean colour}$$

$$\text{or} \quad f_R - f_V = \omega \times f$$

49. **Blue colour of the sky.** According to *Rayleigh's law of scattering*, the intensity of light of wavelength λ present in the scattered light is inversely proportional to the fourth power of wavelength : $I \propto \frac{1}{\lambda^4}$

So, blue colour of sunlight is scattered more by the atmospheric molecules, due to which the sky appears blue.

50. **Rainbow.** It is nature's most spectacular display of the spectrum of light produced by refraction, dispersion and total internal reflection of sunlight by several raindrops. It is observed when the sun shines on rain drops after a shower. An observer standing with his back towards the sun observes it in the form of concentric circular arcs of different colours in the horizon.

Primary rainbow is brighter with its inner edge violet and outer edge red, subtending $41^\circ - 43^\circ$ angle at the observer's eye. Secondary rainbow is fainter with its inner edge red and outer edge violet, subtending $51^\circ - 54^\circ$ angle at the observer's eye.

51. **Human eye.** It is most important and sensitive sense organ. The essential parts of a human eye are sclerotic, cornea, choroid, iris, pupil, crystalline lens, ciliary muscles, aqueous humour, vitreous humour and retina. It is a convex lens of focal length about 2.5 cm.
52. **Accommodation.** It is the ability of the eyelens due to which it can change its focal length so that images of objects at various distances can be formed on the same retina.
53. **Range of normal vision.** The distance between infinity and 25 cm point is called the range of normal vision.
54. **Least distance of distinct vision (D).** The minimum distance from the eye, at which the eye can see the objects clearly and distinctly without any strain is called the least distance of distinct vision. For a normal eye, its value is 25 cm.
55. **Near point.** The nearest point from the eye, at which an object can be seen clearly by the eye is called its near point. The near point of a normal eye is at a distance of 25 cm.
56. **Far point.** The farthest point from the eye, at which an object can be seen clearly by the eye is called the far point of the eye. For a normal eye, the far point is at infinity.
57. **Power of accommodation.** The power of accommodation of the eye is the maximum variation of its power for focussing on near and far objects. For a normal eye, the power of accommodation is about 4 dioptres.
58. **Persistence of vision.** The phenomenon of the continuation of the impression of an image on the retina for some time even after the light from the object is cut off is called persistence of vision. The impression of the image remains on the retina for about $(1/16)$ th of a second. Cinematography works on the principle of persistence of vision.
59. **Rods.** These are rod-shaped cells of the retina that are sensitive to the intensity of light.
60. **Cones.** These are cone-shaped cells of the retina that are sensitive to the colours of light.
61. **Colour blindness.** A person who cannot distinguish between various colours but can see well otherwise, is said to be colour-blind. It is due to lack of some cones in the retina of the eyes.
62. **Cataract.** It is due to the development of hazy or opaque membrane over the eyelens which results in the decrease or loss of vision. It can be cured by surgery.
63. **Common defects of vision.** There are mainly four common defects of vision which can be corrected by the use of suitable eye glasses. These are (i) myopia or near sightedness (ii) hypermetropia or far-sightedness (iii) presbyopia (iv) astigmatism.
64. **Myopia or short-sightedness.** In this defect a person can see nearby objects clearly but cannot see far off objects clearly. Here, either the eyeball becomes too longer or the focal length of the eyelens becomes too short. It can be corrected by using a concave lens of suitable focal length.
- Focal length of the correcting lens
= Distance of the far point from the eye.
65. **Long-sightedness or hypermetropia.** In this defect a person can see the far off objects clearly but he cannot see nearby objects distinctly. Here, either the eyeball becomes too short or the focal length of the eyelens becomes too large. It can be corrected by using convex lens of suitable focal length.
- Focal length of correcting lens = $\frac{yD}{y - D}$
- where y = distance of the near point from the defective eye.
66. **Presbyopia.** In this defect, a person in old age cannot read correctly due to the stiffening of the ciliary muscles and the decrease in flexibility of the eyelens.
67. **Astigmatism.** It is defect of vision in which a person cannot simultaneously see both the horizontal and vertical views of an object with the same clarity. It is due to the irregular curvature of the cornea. It can be corrected by using a cylindrical lens.
68. **Simple microscope.** It is a convex lens of short focal length. When the object is placed between the lens and its focus and the eye is held just behind the lens, a virtual, erect and enlarged image is seen. When the final image is formed at the least distance of distinct vision (D), the magnifying power of the simple microscope is
- $$m = \frac{\text{Angle subtended by the image at the least distance of distinct vision}}{\text{Angle subtended by the object at the least distance of distinct vision}}$$

$$\text{or } m = \frac{\beta}{\alpha} = 1 + \frac{D}{f}$$

When the final image is formed at infinity, $m = \frac{D}{f}$, viewing is more comfortable when the eye is focussed at infinity.

69. **Visual angle.** The angle subtended by an object on the eye is called visual angle. Larger the visual angle, larger is the apparent size of an object.
70. **Compound microscope.** It is an optical device used to see magnified images of tiny objects. The objective is a convex lens of very short focal length and of small aperture. The eyepiece is a convex lens of relatively larger focal length and of larger aperture. The difference between the focal lengths of the eyepiece and the objective is small. Its magnifying power is given by

$$m = m_o \times m_e$$

When the final image is formed at the least distance of distinct vision,

$$m = \frac{\text{Angle subtended by final virtual image at distance } D \text{ from the eye}}{\text{Angle subtended by the object at distance } D \text{ from the eye}}$$

$$\text{or } m = \frac{\beta}{\alpha} = \frac{v_o}{u_o} \left(1 + \frac{D}{f_e} \right) = \frac{L}{f_o} \left(1 + \frac{D}{f_e} \right)$$

When the final image is formed at infinity,

$$m = \frac{L}{f_o} \times \frac{D}{f_e}$$

where L is the distance between the objective and the eyepiece.

71. **Astronomical telescope.** It is used to view heavenly bodies. The objective is a convex lens of large focal length and large aperture. The eyepiece is convex lens of small focal length and small aperture. The difference in the focal lengths of the two lenses is large. The eyepiece forms a real, inverted and diminished image. The eyepiece magnifies this image. The final image is inverted w.r.t. the object.

When the final image is formed at the least distance of distinct vision,

$$m = \frac{\text{Angle subtended by the image at distance } D \text{ from the eye}}{\text{Angle subtended by the object at infinity}}$$

$$\text{or } m = \frac{\beta}{\alpha} = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

$$\text{Length of telescope, } L = f_o + u_e = \frac{f_e D}{f_e + D}$$

When the final image is formed at infinity (normal adjustment),

$$m = \frac{\text{Angle subtended by the final image formed at } \infty}{\text{Angle subtended by the object at } \infty} \quad \text{or } m = \frac{\beta}{\alpha} = -\frac{f_o}{f_e}$$

Length of the telescope in normal adjustment,

$$L = f_o + f_e$$

For large magnifying power of a telescope, clearly

$$f_o \gg f_e$$

72. **Terrestrial telescope.** It is used to see the erect images of distant earthly objects. It uses an additional convex lens between the objective and the eyepiece for erecting the image.

When the final image is formed at infinity, its magnifying power, $m = \frac{f_o}{f_e}$

$$\text{Length of telescope} = f_o + 4f + f_e$$

where f is the focal length of the erecting lens.

When the final image is formed at the least distance of distinct vision, $m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$.

73. **Galileo's telescope.** It uses a concave lens for the eyepiece to obtain an erect image of the distant object. The real, inverted and diminished image formed by the objective lies at the focus of the eyepiece. The final image is formed at infinity and is erect and magnified.

$$\text{In normal adjustment, } m = \frac{f_o}{f_e}$$

$$\text{Length of telescope, } L = f_o - f_e$$

74. **Reflecting telescope.** It uses a concave paraboloidal mirror of large aperture to view the distant objects. Both spherical and chromatic aberrations are minimum.

When the final image is formed at the least distance of distinct vision,

$$m = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

When the final image is formed at infinity,

$$m = \frac{f_o}{f_e} = \frac{R/2}{f_e}$$

COMPETITION SECTION

Wave Optics

GLIMPSES

1. **Nature of light.** The phenomena like interference, diffraction and polarisation establish the wave nature of light. However, the phenomena like black body radiation and photoelectric effect establish the particle nature of light. de Broglie suggested that *light has a dual nature i.e., it can behave as particles as well as waves.*
2. **Wavefront.** A wavefront is defined as the continuous locus of all such particles of the medium which are vibrating in the same phase at any instant. In case of waves travelling in all directions from a point source, the wavefronts are spherical in shape. When the source of light is linear in shape, the wavefronts are cylindrical. At very large distances from the source, a portion of spherical or cylindrical wavefront is plane wavefront.
3. **Ray.** An arrow drawn perpendicular to a wavefront in the direction of propagation of a wave is called a ray.
Two general principles are valid for rays and wavefronts :
 - (i) Rays are normal to wavefronts.
 - (ii) The time taken to travel from one wavefront to another is the same along any ray.
4. **Huygens' principle of secondary wavelets.** Huygens' principle is the basis of the wave theory of light. It tells how a wavefront propagates through a medium. It is based on the following assumptions :
 - (i) Each point on a wavefront acts as a source of new disturbance called secondary waves or wavelets.
 - (ii) The secondary wavelets spread out in all directions with the speed of light in the given medium.
 - (iii) The wavefront at any later time is given by the forward envelope of the secondary wavelets at that time.
5. **Effect on frequency, wavelength and speed during refraction.** When a light wave travels from one medium to another, its frequency remains unchanged but both its wavelength and speed get changed, depending on the refractive index of the refracting medium.
6. **Interference of light waves.** When light waves from two coherent sources travelling in the same direction superpose each other, the intensity in the region of superposition gets redistributed, becoming maximum at some points and minimum at others. This phenomenon is called interference of light.
7. **Constructive and destructive interference.** If path difference $p = n\lambda$ or phase difference $\phi = 2n\pi$, the two waves are in same phase and so add up to give maximum of intensity. This is called *constructive interference*.
If $p = (2n - 1)\lambda/2$ or $\phi = (2n - 1)\pi$, the two superposing waves are out of phase, the resultant amplitude is equal to difference between their individual amplitudes and hence intensity is minimum. This is called *destructive interference*.
8. **Young's double slit experiment.** In Young's double slit experiment, two identical narrow slits S_1 and S_2 are placed symmetrically with respect to narrow slit S illuminated with

monochromatic light. The interference pattern is obtained on an observation screen placed at large distance D from S_1 and S_2 .

The position of n th bright fringe from the centre of screen is

$$x_n = n \frac{D\lambda}{d}$$

The position of n th dark fringe from the centre of the screen is

$$x'_n = (2n-1) \frac{D\lambda}{2d}$$

Fringe width is the separation between two successive bright or dark fringes and is given by

$$\beta = \frac{D\lambda}{d}$$

9. **Resultant amplitude and intensity of interfering waves.** If a_1 and a_2 are the amplitudes and I_1 and I_2 are the intensities of two coherent waves having phase difference ϕ , then their resultant amplitude and intensity at the point of superposition are given by

$$a = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$$

$$\text{and } I = I_1 + I_2 + 2\sqrt{I_1I_2} \cos \phi$$

If amplitude of each wave is a_0 and intensity I_0 , then

$$\begin{aligned} I &= 2ka_0^2 (1 + \cos \phi) = 2I_0 (1 + \cos \phi) \\ &= 4I_0 \cos^2 \frac{\phi}{2} \end{aligned}$$

The term $2\sqrt{I_1I_2} \cos \phi$ is called *interference term*.

- (i) When $\cos \phi$ remains constant with time, the two sources are coherent. The intensity will be maximum at points for which $\cos \phi = +1$ and minimum at points for which $\cos \phi = -1$.
 - (ii) When $\cos \phi$ varies continuously with time so that its average value is zero over the time interval of measurement, the resultant intensity at all points will be $I_1 + I_2$. No interference fringes are observed. The sources are incoherent.
10. **Ratio of intensity at maxima and minima of an interference pattern.** If a_1 and a_2 are the amplitudes of two interfering waves, then the ratio between the intensities at maxima and minima will be

$$\frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \left(\frac{r+1}{r-1} \right)^2$$

where $r = \frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}}$, is the amplitude ratio of

two waves. If w_1 and w_2 are the widths of the two slits, then

$$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2}$$

11. **Coherent sources.** Two sources of light which continuously emit light waves of same frequency (or wavelength) with a zero or constant phase difference between them, are called coherent sources. Two independent sources of light cannot act as coherent sources, they have to be derived from the same parent source.
12. **Conditions for sustained interference :**
 - (i) The two sources should continuously emit waves of same frequency or wavelength.
 - (ii) The two sources of light should be coherent.
 - (iii) The amplitudes of the interfering waves should be equal.
 - (iv) The two sources should be narrow.
 - (v) The interfering waves must travel nearly along the same direction.
 - (vi) The sources should be monochromatic.
 - (vii) The interfering waves should be in the same state of polarisation.
 - (viii) The distance between the two coherent sources should be small and the distance between the two sources and the screen should be large.
13. **Fresnel's biprism method.** Here two coherent sources are obtained from an incoherent source, by refraction. A biprism is essentially a single prism with an obtuse angle of 179° , but behaves as a combination of two acute angled prisms placed base to base, each with a refracting angle of about $\frac{1^\circ}{2}$.
14. **Lloyd's single mirror method.** In this method, an illuminated slit and its reflected image serve as two coherent sources. In contrast to Young's double slit and Fresnel's biprism methods, here the central fringe is dark.

15. **Displacement of interference fringes.** When a thin transparent sheet of thickness t and refractive index μ is inserted in the path of one of the interfering beams, the extra path difference introduced is

$$\Delta p = \text{Length } t \text{ in transparent sheet} \\ - \text{Length } t \text{ in air}$$

$$\text{or } \Delta p = \mu t - t = (\mu - 1)t$$

$$\therefore \text{Net path difference for any point on the screen} \\ = \frac{xd}{D} - (\mu - 1)t$$

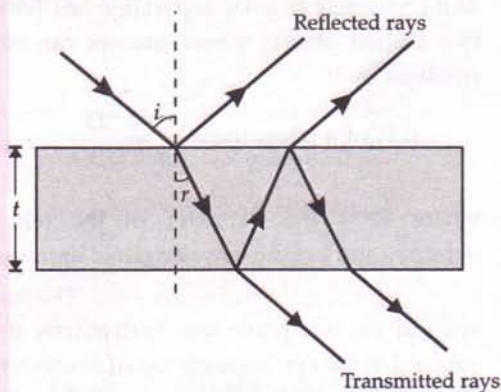
For the central point of the screen,

$$\frac{xd}{D} - (\mu - 1)t = 0 \quad \text{or} \quad x = \frac{D}{d}(\mu - 1)t$$

Thus the shift in the central bright fringe and hence shift of any other fringe is

$$\Delta x = \frac{D}{d}(\mu - 1)t = \frac{\beta}{\lambda}(\mu - 1)t$$

16. **Interference in thin films.** A soap film or thin film of oil spread over water shows beautiful colours, when seen in the reflected light. This is due to interference between light waves reflected by the upper and lower surfaces of thin films, as shown in the figure below. The ray reflected from the upper denser surface of thin film suffers a phase change of π or path difference of $\lambda/2$.



Reflected system. The path difference between the two consecutive rays reflected from the upper and the lower surfaces of a thin film of refractive index μ and thickness t is given by

$$p = 2\mu t \cos r - \frac{\lambda}{2}$$

$$\text{For maximum intensity. } 2\mu t \cos r = (2n + 1)\frac{\lambda}{2}$$

$$\text{For minimum intensity. } 2\mu t \cos r = n\lambda$$

Transmitted system.

$$\text{For maximum intensity. } 2\mu t \cos r = n\lambda$$

For minimum intensity.

$$2\mu t \cos r = (2n + 1)\frac{\lambda}{2}, \quad \text{where } n = 0, 1, 2, 3, \dots$$

17. **Diffraction of light.** The phenomenon of bending of light around the corners of small obstacles or apertures and their consequent spreading into the regions of geometrical shadow is called diffraction of light.

18. **Diffraction at a single slit.** A plane wave of wavelength λ on passing through a narrow slit of width a suffers diffraction producing a central bright fringe ($\theta = 0^\circ$), flanked on both sides by minima and maxima. The intensity of secondary maxima decreases with the increase in distance from the centre.

For n th minimum :

$$a \sin \theta_n = n\lambda, \quad n = 1, 2, 3, \dots$$

For n th secondary maximum :

$$a \sin \theta_n = (2n + 1)\frac{\lambda}{2}, \quad n = 1, 2, 3, \dots$$

Angular position of n th minimum,

$$\theta_n = \frac{n\lambda}{a}$$

Distance of n th minimum from the centre of the screen,

$$x_n = n \frac{D\lambda}{a}$$

Angular position of n th secondary maximum,

$$\theta'_n = (2n + 1)\frac{\lambda}{2a}$$

Distance of n th secondary maximum from the centre of the screen,

$$x'_n = (2n + 1)\frac{D\lambda}{2a}$$

Width of a secondary maximum,

$$\beta = \frac{D\lambda}{a}$$

Width of central maximum,

$$\beta_0 = 2\beta = \frac{2D\lambda}{a}$$

Angular spread of central maximum on either side of the centre of the screen is

$$\theta = \pm \frac{\lambda}{a}$$

Total angular spread of the central maximum is

$$2\theta = \frac{2\lambda}{a}$$

For diffraction to be more pronounced, the size of the slit should be comparable to the wavelength of light used.

19. **Diffraction at a circular aperture.** For diffraction of light at a circular aperture of diameter a , the angular spread of central maximum is

$$\theta = \frac{1.22\lambda}{a}$$

If D is the distance at which the effect is observed, then

Linear spread, $x = D\theta$

Areal spread, $x^2 = (D\theta)^2$.

20. **Fresnel's distance.** It is the distance at which the diffraction spread of a beam becomes equal to the size of the aperture. If a is the width of the aperture, then

$$D_F = \frac{a^2}{\lambda}$$

The ray optics is valid for a distance $D < D_F$.

21. **Diffraction grating.** It is an arrangement of a very large number of very narrow, equidistant and parallel slits. The diffraction pattern has the central principal maximum of maximum intensity and a number of higher order intensity maxima whose intensity decreases with the increase of n , the order of the spectrum. The direction of n th principal maximum is given by

$$(a + b) \sin \theta_n = n\lambda$$

where $n = 0, 1, 2, 3, \dots$

This equation is known as *grating law*. Here $(a + b)$ is called *grating element*, where a is width of each slit and b is the width of opaque space between two consecutive slits.

22. **Limit of resolution.** The smallest linear or angular separation between two point objects at which they can be just resolved by an optical instrument is called the limit of resolution of the instrument.
23. **Resolving power.** It is the ability of an optical instrument to resolve or separate the images of two nearby point objects so that they can be

distinctly seen. It is equal to the reciprocal of the limit of resolution of the optical instrument.

24. **Diffraction as a limit on resolving power.** All optical instruments like lens, telescope, microscope, etc. act as apertures. Light on passing through them undergoes diffraction. This puts the limit on their resolving power.
25. **Rayleigh's criterion for resolution.** The images of two point objects are just resolved when the central maximum of the diffraction pattern of one falls over the first minimum of the diffraction pattern of the other.
26. **Resolving power of a microscope.** The resolving power of a microscope is defined as the reciprocal of the smallest distance d between two point objects at which they can be just resolved when seen in the microscope.

$$\text{R.P. of a microscope} = \frac{1}{d} = \frac{2\mu \sin \theta}{\lambda}$$

where θ is half the angle of cone of light from each point object and μ is the refractive index of the medium between the object and the objective.

The factor $\mu \sin \theta$ is called *numerical aperture* (N.A.).

27. **Resolving power of a telescope.** The resolving power of a telescope is defined as the reciprocal of the smallest angular separation ' $d\theta$ ' between two distant objects whose images can be just resolved by it.

$$\text{R.P. of a telescope} = \frac{1}{d\theta} = \frac{D}{122\lambda}$$

where D is the diameter of the telescope objective and λ is the wavelength of light used.

28. **Resolving power of the human eye.** The human eye can see two point objects distinctly if they subtend at the eye, an angle equal to one minute of arc. This angle is called the limit of resolution of the eye. The reciprocal of this angle equals the resolving power of the eye.
29. **Polarisation of waves.** A transverse wave in which vibrations are present in all possible directions, in a plane perpendicular to the direction of propagation, is said to be unpolarised. If the vibrations of a wave are present in just one direction in a plane perpendicular to the

direction of propagation, the wave is said to be polarised or plane polarised. The phenomenon of restricting the oscillations of a wave to just one direction in the transverse plane is called polarisation.

30. **Unpolarised light.** A kind of light in which the electric field vector takes all possible directions in the transverse plane, rapidly and randomly, during the time of measurement is called unpolarised light. For example, the light of the sun, candle light, etc.
31. **Plane polarised light.** If the electric field vector vibrates just in one direction perpendicular to the direction of wave propagation, the light is said to be linearly polarised. In a linearly polarised wave, the vibrations at all points, at all times, lie in the same plane, so it is also called a plane polarised wave.
32. **Polariser.** A device that plane polarises the unpolarised light passed through it is called a polariser. For example, a tourmaline crystal, nicol prism, polaroid, etc.
33. **Law of Malus.** This law states that when a beam of completely plane polarised light is passed through an analyser, the intensity ' I ' of the transmitted light varies directly as the square of the angle ' θ ' between the transmission directions of polariser and analyser.

$$I = I_0 \cos^2 \theta$$

where I_0 is the maximum intensity of transmitted light.

34. **Plane of polarisation.** The plane passing through the direction of wave propagation and perpendicular to the plane of vibration is called the plane of polarisation.
35. **Plane of vibration.** The plane containing the direction of vibration and the direction of wave propagation is called the plane of vibration.
36. **Brewster angle.** The angle of incidence at which a beam of unpolarised light falling on a transparent surface is reflected as a beam of completely plane polarised light is called polarising or Brewster angle. It is denoted by i_p .
37. **Brewster law.** This law states that the tangent of the polarising angle of incidence of a transparent medium is equal to its refractive index.

$$\mu = \tan i_p$$

38. **Nicol prism.** It is an optical device based on the phenomenon of double refraction which is used for producing and analysing plane polarised light. It consists of two pieces of calcite cut with a 68° angle and stuck together with Canada balsam.
39. **Polaroids.** These are thin commercial sheets which make use of the property of selective absorption (dichroism) to produce an intense beam of plane polarised light. Polaroids are used in sunglasses, camera filters, wind screens and car head lights of motor cars to reduce glare of light reflected from shiny surfaces, etc.
40. **Optical activity.** Substances which can rotate the plane of polarisation of light are called optically active substances while the phenomenon is called optical activity.
41. **Specific rotation.** It is the angle through which the plane of polarisation rotates when plane polarised light is passed through one decimetre length of solution containing one gram of the substance per cm^3 . The measurement is done at a given temperature T , using sodium light (the D-line).

Specific rotation

$$= \frac{\text{Observed angle of rotation in degrees}}{\text{Length of the tube in decimetre} \times \text{Grams of substance in } 1 \text{ cm}^3 \text{ of solution}}$$

$$[s]_D^T = \frac{\theta}{l \times c}$$

42. **Doppler effect.** It is the phenomenon of the apparent change in the frequency of light due to the relative motion between the source and observer. The apparent frequency ν' is given by

$$\nu' = \nu \left(1 \pm \frac{v}{c} \right)$$

When source moves towards the observer, velocity v is taken $+ve$ and when it moves away from the observer, v is taken $-ve$.

43. **Doppler shift.** The apparent change in the frequency of light due to Doppler effect is called Doppler shift.

$$(i) \Delta \nu = \pm \frac{v}{c} \nu$$

$$(ii) \Delta \lambda = \mp \frac{v}{c} \cdot \lambda.$$

COMPETITION SECTION

Dual Nature of Radiation and Matter

GLIMPSES

1. **Electron.** It is an elementary particle having a negative charge of 1.6×10^{-19} C and mass 9.1×10^{-31} kg.
2. **Work function.** The minimum amount of energy required by an electron to just escape from the metal surface is known as work function of the metal. It is denoted by W_0 .
3. **Electron emission.** The phenomenon of emission of electrons from a metal surface is called electron emission. It is of the following types :
 - (i) **Thermionic emission.** Here electrons are emitted from the metal surface with the help of thermal energy.
 - (ii) **Field or cold cathode emission.** Electrons are emitted from a metal surface by subjecting it to a very high electric field.
 - (iii) **Photoelectric emission.** Electrons are emitted from a metal surface with the help of suitable electromagnetic radiations.
 - (iv) **Secondary emission.** Electrons are ejected from a metal surface by striking fast moving electrons over it.
4. **Kinetic energy gained by an electron.** When an electron is accelerated from rest through a potential difference of V volts, the gain in its kinetic energy is

$$eV = \frac{1}{2}mv^2$$

5. **Electron volt (eV).** It is the kinetic energy gained by an electron when it is accelerated through a potential difference of 1 volt.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \quad 1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$$

The work function of a metal is generally measured in electron volt (eV).

6. **Photons.** According to Planck's quantum theory of radiation, an electromagnetic wave travels in the form of discrete packets of energy called quanta. One quantum of light radiation is called a photon. The main features of photons are as follows :

- (i) A photon travels with the speed of light.
- (ii) The frequency of a photon does not change as it travels from one medium to another.
- (iii) The speed of a photon changes as it travels through different media due to the change in its wavelength.
- (iv) The rest mass of a photon is zero i.e., a photon cannot exist at rest.
- (v) Energy of a photon, $E = h\nu = \frac{hc}{\lambda}$.
- (vi) Momentum of a photon, $p = mc = \frac{h\nu}{c} = \frac{h}{\lambda}$.
- (vii) From Einstein's mass-energy relationship, the equivalent mass m of a photon is given by

$$E = mc^2 = h\nu \quad \text{or} \quad m = \frac{h\nu}{c^2}$$

7. **Photoelectric effect.** The phenomenon of emission of electrons from a metal surface, when electromagnetic radiations of sufficiently high frequency are incident on it, is called photoelectric effect. The photo (light)-generated electrons are called photoelectrons.

Alkali metals like Li, Na, K, Ce show photoelectric effect with visible light. Metals like Zn, Cd, Mg respond to ultraviolet light.

Photoelectric effect involves the conversion of light energy into electrical energy. It follows the law of conservation of energy. It is an instantaneous process.

8. **Photoelectric current.** The current constituted by photoelectrons is called photoelectric current. Its value depends on :

- (i) the intensity of light,
- (ii) the potential difference applied between the two electrodes, and
- (iii) the nature of the cathode material.

9. **Cut off or stopping potential.** It is the minimum value of the negative potential that must be applied to the anode of photo-cell to make the photoelectric current zero. It is denoted by V_0 . Its value depends on : (i) the frequency of incident light, and (ii) the nature of the cathode material. For a given frequency of incident light, it is independent of its intensity. The stopping potential is directly related to the kinetic energy of the emitted electrons.

$$K_{\max} = \frac{1}{2} m v_{\max}^2 = eV_0$$

10. **Threshold frequency.** The minimum value of the frequency of incident radiation below which the photoelectric emission stops altogether is called threshold frequency. It is denoted by ν_0 and is a characteristic of the metal.

11. **Laws of photoelectric effect.** (i) For a given metal and a radiation of fixed frequency, the rate of emission of photoelectrons is proportional to the intensity of incident radiation. (ii) For every metal, there is a certain minimum frequency below which no photoelectrons are emitted, howsoever high is the intensity of incident radiation. This frequency is called *threshold frequency*. (iii) For the radiation of frequency higher than the threshold frequency, the maximum kinetic energy of the photoelectrons is directly proportional to the frequency of incident radiation and is independent of the intensity of incident radiation. (iv) The photoelectric emission is an *instantaneous process*.

12. **Failure of wave theory to explain photoelectric effect.** The classical wave theory of radiation could not explain the main features of photoelectric effect. Its picture of continuous absorption of energy from radiation could not explain (i) the independence of $K_{\max}$ on intensity, (ii) the existence of threshold frequency ν_0 and (iii) the instantaneous nature of the phenomenon.

13. **Einstein's theory of photoelectric effect.** Einstein explained photoelectric effect with the help of Planck's quantum theory. When a radiation of frequency ν is incident on a metal surface, it is absorbed in the form of discrete packets of energy called quanta or photons. A part of energy $h\nu$ of a photon is used in removing the electron from the metal surface and remaining energy is used in giving kinetic energy to the photoelectron. Einstein's photoelectric equation is

$$K_{\max} = \frac{1}{2} m v_{\max}^2 = eV_0 = h\nu - W_0 = h(\nu - \nu_0)$$

where W_0 is the work function of the metal and ν_0 is the threshold frequency.

All the experimental observations can be explained on the basis of Einstein's photoelectric equation.

14. **Compton scattering.** It is the phenomenon of increase in the wavelength of X-ray photons which occurs when these radiations are scattered on striking an electron. Then difference in the wavelength of scattered and incident photons is called *Compton shift*, which is given by

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos \phi)$$

where ϕ is the angle of scattering of the X-ray photon and m_0 is the rest mass of the electron.

15. **Photocell.** It is an arrangement which converts light energy into electric energy. It works on the principle of photoelectric effect. It is used in cinematography for the reproduction of sound.

Photo-cells are used to operate various control systems and in light measuring devices.

16. **Dual nature of radiation.** Light has dual nature. It manifests itself as a wave in diffraction, interference, polarisation, etc., while it shows particle nature in photoelectric effect, Compton scattering, etc.

17. **Dual nature of matter.** As there is complete equivalence between matter (mass) and radiation (energy) and the principle of symmetry is always obeyed, de Broglie suggested that moving particles like protons, neutrons, electrons, etc. should be associated with waves known as *de Broglie waves* and their wavelength is called *de Broglie wavelength*. The de Broglie

wavelength of a particle of mass m moving with velocity v is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where h is the Planck's constant. The de Broglie wavelength is independent of the charge and nature of the material particles. It has significantly measurable values for sub-atomic particles like electrons, protons, etc., due to their small masses. For macroscopic objects of everyday life, the de-Broglie wavelength is extremely small, quite beyond measurement.

18. **Davisson and Germer experiment.** This electron diffraction experiment has verified and confirmed the wave-nature of electrons.

19. **de Broglie wavelength of an electron.** The wavelength associated with an electron beam accelerated through a potential difference of V volts is given by

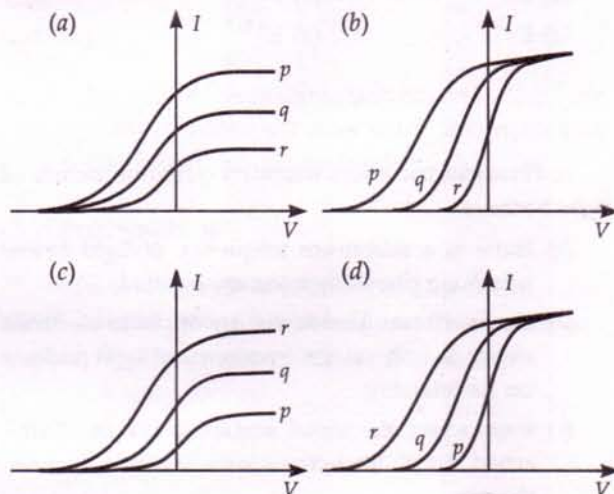
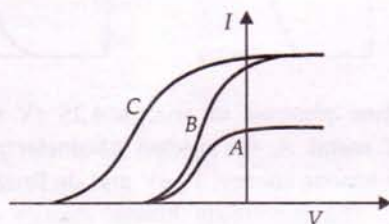
$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{1.227}{\sqrt{V}} \text{ nm}$$

20. **Electron microscope.** It is a device that exploits the wave-nature of electrons to provide high resolving power. It is used to investigate the structural details of bacteria, viruses, etc. It has proved to be a powerful tool of investigation for research in science, technology, metallurgy, industry, medicine, etc.

JEE Advance

Multiple Choice Questions with one correct answer

1. In a photoelectric experiment anode potential is plotted against plate current.



- (a) A and B will have same intensities while B and C will have different frequencies
(b) B and C will have different intensities while A and B will have different frequencies
(c) A and B will have different intensities while B and C will have equal frequencies
(d) B and C will have equal intensities while A and B will have same frequencies. [IIT 2004]

2. Photoelectric effect experiments are performed using three different metal plates p , q and r having work functions $\phi_p = 2.0 \text{ eV}$, $\phi_q = 2.5 \text{ eV}$ and $\phi_r = 3.0 \text{ eV}$, respectively. A light beam containing wavelengths of 550 nm , 450 nm and 350 nm with equal intensities illuminates each of the plates. The correct I - V graph for the experiment is [IIT 2009]

3. The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 4 eV . The stopping potential, in volt, is

- (a) 2 (b) 4
(c) 6 (d) 10 [IIT 1997]

4. A metal surface is illuminated by light of two different wavelengths 248 nm and 310 nm . The maximum speeds of the photoelectrons corresponding to these wavelengths are u_1 and u_2 , respectively. If the ratio $u_1 : u_2 = 2 : 1$ and $hc = 1240 \text{ eV nm}$, the work function of the metal is nearly

- (a) 3.7 eV (b) 3.2 eV
(c) 2.8 eV (d) 2.5 eV [JEE Adv. 14]

COMPETITION SECTIONS

Atoms

GLIMPSES

1. **Atom as a constituent of every element.** Every element has characteristic atoms. Atoms of different elements contain electrons, which are completely identical. Atom as a whole is electrically neutral and therefore contains equal amount of positive and negative charges.

2. **Thomson's model of an atom.** An atom consists of a sphere of positively charged matter in which the negatively charged electrons are uniformly embedded like plums in a pudding. This model could not explain scattering of α -particles through thin foils and hence discarded.

3. **Rutherford's model of an atom.** Geiger and Marsden performed experiments on scattering of α -particles from metal foils. A collimated beam of 5.5-MeV of α -particles was allowed to fall on a 2.1×10^{-7} m thin gold foil. The scattered α -particles produced scintillations on a ZnS screen, which were counted at different angles (θ) from the direction of the beam. It was found that most of the α -particles passed undeviated through thin foils but some of them were scattered through very large angles. From the results of these experiments, Rutherford proposed the following model of an atom.

(i) An atom consists of a small and massive central core in which the entire positive charge and almost the whole mass of the atom are concentrated. This core is called the **nucleus**.

(ii) The nucleus occupies a very small space as compared to the size of the atom.

(iii) The atom is surrounded by a suitable number of electrons so that their total negative charge is equal to the total positive charge on the nucleus and the atom as a whole is electrically neutral.

(iv) The electrons revolve around the nucleus in various orbits just as planets revolve around the sun. The centripetal force required for their revolution is provided by the electrostatic attraction between the electrons and the nucleus.

This model could not explain the stability of the atom because according to classical electromagnetic theory the electron revolving around the nucleus must continuously radiate energy in the form of electromagnetic radiations and hence it should fall into the nucleus.

4. **Distance of closest approach.** When an α -particle of mass m and velocity v moves directly towards a nucleus of atomic number Z , its distance of closest approach is given by

$$r_0 = \frac{2kZe^2}{E} = \frac{4kZe^2}{mv^2}$$

where $E = \frac{1}{2}mv^2$

and $k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$.

5. **Impact parameter.** It is defined as the perpendicular distance of the velocity vector of the α -particle from the centre of the nucleus, when it is far away from the atom. The shape of the trajectory of the scattered α -particle depends on the impact parameter b and the nature of the

potential field. Rutherford deduced the following relationship between the impact parameter b and the scattering angle θ :

$$b = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2 \cot \frac{\theta}{2}}{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2 \cot \frac{\theta}{2}}{\frac{1}{2}mv^2}.$$

6. **Quantisation or discretisation.** The quantisation or discretisation of a physical quantity means that it cannot vary continuously to have any arbitrary value but can change only discontinuously to take certain specific values. The energy of electrons in an atom is quantised.

7. **Bohr's atom model.** This model is also called *planetary model* of an atom and is based on following postulates :

(i) **Nuclear concept.** An atom consists of a small massive central core called nucleus around which planetary electrons revolve. The centripetal force required for their revolution is provided by the electrostatic attraction between the electrons and the nucleus.

(ii) **Quantum condition.** Of all the possible circular orbits allowed by the classical theory, the electrons are permitted to circulate only in such orbits in which the angular momentum of an electron is an integral multiple of $h/2\pi$, h being Planck's constant.

$$L = mvr = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

where n is called *principal quantum number*.

(iii) **Stationary orbits.** While revolving in the permissible orbits, an electron does not radiate energy. These non-radiating orbits are called stationary orbits.

(iv) **Frequency condition.** An atom can emit or absorb radiation in the form of discrete energy photons only when an electron jumps from a higher to a lower orbit or from a lower to a higher orbit. If E_1 and E_2 are the energies associated with these permitted orbits, then the frequency ν of the emitted or absorbed radiation is given by

$$h\nu = E_2 - E_1$$

8. **Bohr's theory of hydrogen atom.** An electron having charge $-e$ revolves with speed v in a circular orbit of radius r round the nucleus having charge $+e$.

For a circular orbit,

$$\frac{mv^2}{r} = \frac{kZe^2}{r}$$

From quantisation of angular momentum,

$$L = mvr = \frac{nh}{2\pi}$$

On solving the above two equations, we get

Radius of n th orbit,

$$r = \frac{n^2 h^2}{4\pi^2 m k Z e^2}$$

Speed of electron in n th orbit,

$$v = \frac{2\pi k e^2}{nh} = \alpha \frac{c}{n} = \frac{1}{137} \cdot \frac{c}{n}.$$

where $\alpha = \frac{2\pi k e^2}{ch}$ is *fine structure constant*.

Total energy of an electron in n th orbit is

$$E_n = \text{K.E.} + \text{P.E.} = \frac{kZe^2}{2r} - \frac{kZe^2}{r} = -\frac{kZe^2}{2r}$$

$$\begin{aligned} \text{or } E_n &= -\frac{2\pi^2 m k^2 Z^2 e^4}{n^2 h^2} \\ &= -\frac{Z^2 R h c}{n^2} = -\frac{13.6 Z^2}{n^2} \text{ eV} \end{aligned}$$

$$\begin{aligned} \text{where } R &= \frac{2\pi^2 m k^2 e^4}{ch^3} \\ &= 1.0973 \times 10^7 \text{ m}^{-1}, \end{aligned}$$

is the *Rydberg's constant*.

9. **Spectral series of hydrogen atom.** Whenever an electron makes a transition from a higher energy level n_2 to a lower energy level n_1 , the difference of energy appears in the form of a photon of frequency ν given by

$$\nu = \frac{2\pi^2 m k^2 Z^2 e^4}{h^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Wave number,

$$\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} = \frac{2\pi^2 m k^2 Z^2 e^4}{ch^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\text{or } \bar{\nu} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Different spectral series for hydrogen atom are as follows :

- (i) **Lyman Series.** Here $n_2 = 2, 3, 4, \dots$ and $n_1 = 1$. This series lies in the *ultraviolet region*.

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{n_2^2} \right], \quad n_2 = 2, 3, 4, \dots$$

- (ii) **Balmer Series.** Here $n_2 = 3, 4, 5, \dots$ and $n_1 = 2$. This series lies in the *visible region*.

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{2^2} - \frac{1}{n_2^2} \right], \quad n_2 = 3, 4, 5, \dots$$

- (iii) **Paschen Series.** Here $n_2 = 4, 5, 6, \dots$ and $n_1 = 3$. This series lies in the *infrared region*.

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{3^2} - \frac{1}{n_2^2} \right], \quad n_2 = 4, 5, 6, \dots$$

- (iv) **Brackett Series.** Here $n_2 = 5, 6, 7, \dots$ and $n_1 = 4$. This series lies in the *infrared region*.

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{4^2} - \frac{1}{n_2^2} \right], \quad n_2 = 5, 6, 7, \dots$$

- (v) **Pfund Series.** Here $n_2 = 6, 7, 8, \dots$ and $n_1 = 5$. This series lies in the *infrared region*.

$$\bar{\nu} = \frac{1}{\lambda} = R \left[\frac{1}{5^2} - \frac{1}{n_2^2} \right], \quad n_2 = 6, 7, 8, \dots$$

10. **Energy level diagram.** It is a diagram in which the energies of the different stationary states of an atom are represented by parallel horizontal lines, drawn according to some suitable energy scale.
11. **Failure of Bohr's model.** This model is applicable only to hydrogen-like atoms and fails in case of higher atoms. It could not explain the fine structure of the spectral lines in the spectrum of hydrogen atom.
12. **Excitation energy.** It is defined as the energy required by an electron of an atom to jump from its ground state to any one of its excited states.
13. **Ionisation energy.** It is defined as the energy required to remove an electron from an atom, i.e., the energy required to take an electron from its ground state to the outermost orbit ($n = \infty$).
14. **Excitation potential.** It is that accelerating potential which gives to a bombarding electron sufficient energy to excite the target atom by raising one of its electrons from an inner to an outer orbit.
15. **Ionisation Potential.** It is that accelerating potential which gives to a bombarding electron sufficient energy to ionise the target atom by knocking one of its electrons completely out of the atom.

COMPETITION SECTIONS

Nuclei

GLIMPSES

1. **Nucleons.** Protons and neutrons which are present in the nuclei of the atoms are collectively known as nucleons.

2. **Atomic Number.** The number of protons present in the nucleus is called the atomic number. It is denoted by Z .

3. **Mass number.** The total number of protons and neutrons present in a nucleus is called the mass number of the element. It is denoted by A .

Number of protons in an atom $= Z$

Number of electrons in an atom $= Z$

Number of nucleons in an atom $= A$

Number of neutrons in an atom $= N = A - Z$.

4. **Nuclear mass.** The total mass of the protons and neutrons present in a nucleus is called the nuclear mass.

5. **Nuclide.** A nuclide is a specific nucleus of an atom characterised by its atomic number Z and mass number A . It is represented as



X = chemical symbol of the element,

Z = atomic number, and

A = mass number.

6. **Isotopes.** The atoms of an element, which have the same atomic number but different mass number are called isotopes.

7. **Isobars.** The atoms of different elements (different atomic number) but having the same mass number are called isobars.

8. **Isotones.** The nuclides having the same number of neutrons are called isotones.

9. **Isomers.** These are the nuclei with same atomic number and same mass number but in different energy states.

10. **Atomic mass unit (amu).** It is defined as $\frac{1}{12}$ th of the mass of one ${}^{12}_6\text{C}$ atom. Its value is given by
 $1 \text{ amu} = 1.660565 \times 10^{-27} \text{ kg} = 931 \text{ MeV}.$

11. **Electron volt.** It is defined as the energy acquired by an electron when it is accelerated through a potential difference of 1 volt and is denoted by eV.

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J},$$

$$1 \text{ MeV} = 10^6 \text{ eV} = 1.602 \times 10^{-13} \text{ J}.$$

12. **Relation between size of nucleus and mass number.** It is found that the radius r of a nucleus is proportional to the cube root of its mass number.

$$r = r_0 A^{1/3},$$

$$\text{where } r_0 = 1.2 \times 10^{-15} \text{ m}$$

$= 1.2 \text{ fermi}$ for electrons as probes.

13. **Nuclear density.** The density of a nucleus is independent of the size of the nucleus and is given by

$$\rho_{\text{nu}} = \frac{\text{Nuclear mass}}{\text{Nuclear volume}} = \frac{m_{\text{nu}}}{\frac{4}{3} \pi R^3}.$$

The nuclear density is of the order of $10^{17} \text{ kg m}^{-3}$.

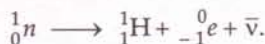
14. **Discovery of neutrons.** Neutrons were discovered by Chadwick in 1932. When beryllium nuclei are bombarded by α -particles, highly penetrating radiations are emitted, which consist of neutral particles, each having mass

(13.77)

nearly that of a proton. These particles were called neutrons.



A free neutron decays spontaneously, with a half life of about 1000 s, into a proton, electron and an antineutrino.



15. **Nuclear forces.** These are the strong attractive forces which hold protons and neutrons together in a tiny nucleus. These are short range forces which operate over very short distances of about 2–3 fm from a nucleon. The nuclear force does not depend on the charge of the nucleus.

16. **Mass defect.** The difference between the rest mass of a nucleus and the sum of the rest masses of its constituent nucleons is called its mass defect. It is given by

$$\Delta m = Zm_p + (A - Z)m_n - m$$

17. **Binding energy.** It may be defined as the energy required to break up a nucleus into its constituent protons and neutrons and to separate them to such a large distance that they may not interact with each other.

It may also be defined as the surplus energy which the nucleons give up by virtue of their attractions when they become bound together to form a nucleus.

The binding energy of a nucleus ${}^A_Z\text{X}$ is given by

$$\Delta E_b = [Zm_H + (A - Z)m_n - m]c^2$$

18. **Binding energy per nucleon.** It is the average energy required to extract one nucleon from the nucleus. It is obtained by dividing the binding energy of a nucleus by its mass number.

$$\Delta E_{bn} \text{ or } \bar{B} = \frac{\text{B.E.}}{A} = \frac{[Zm_H + (A - Z)m_n - m]c^2}{A}$$

In the mass number range $A = 30$ to 170, the binding energy per nucleon is nearly constant, about 8 MeV per nucleon.

19. **Packing fraction.** The packing fraction of a nucleus is its mass defect per nucleon.

$$\text{P.F. of a nucleus} = \frac{\text{Mass defect}}{\text{Mass number}} = \frac{\Delta m}{A}$$

It is directly related to the availability of nuclear energy and the stability of the nucleus.

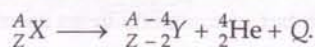
20. **Nuclear energy levels.** Like an atom, the nucleus exists in nucleonic configurations, which correspond to nuclear stationary states. The stationary state of the lowest energy is called the ground state. When a nucleus makes a transition from one level to a level of lower energy, the emitted photon belongs to the gamma ray region of the electromagnetic spectrum.

21. **Radioactivity.** It is the phenomenon of spontaneous disintegration of the nucleus of an atom with the emission of one or more radiations like α -particles, β -particles or γ -rays. The substances which spontaneously emit these penetrating radiations are called radioactive substances.

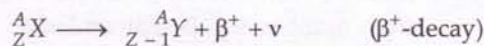
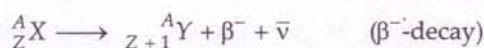
22. **Soddy-Fajan's displacement law.** It states that

- (i) When a radioactive nucleus emits an α -particle, its atomic number decreases by 2 and mass number decreases by 4.
- (ii) When a radioactive nucleus emits β -particle, its atomic number increases by 1 but mass number remains same.
- (iii) The emission of a γ -particle does not change the mass number or the atomic number of the radioactive nucleus.

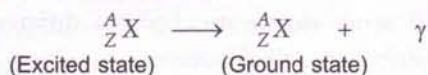
23. **Alpha decay.** It is the process of emission of an α -particle from a radioactive nucleus.



24. **Beta decay.** It is the process of emission of an electron or a positron from a radioactive nucleus.



25. **Gamma decay.** It is the process of emission of a γ -ray photon during the radioactive disintegration of a nucleus. This occurs when a nucleus in an excited state makes a transition to a state of lower energy



26. **Radioactive decay law.** The number of atoms of a radioactive sample disintegrating per second at any instant is directly proportional to the number of undecayed radioactive nuclei present at that instant.

$$\frac{dN}{dt} = -\lambda N$$

The law may also be expressed as $N = N_0 e^{-\lambda t}$ where N_0 is the number of nuclei at $t = 0$ and λ is decay constant.

27. **Decay constant.** It may be defined as the reciprocal of the time interval in which the number of active nuclei in a given radioactive sample reduces to 36.8% (or $1/e$ times) of its initial value. Its units are s^{-1} , day^{-1} , $year^{-1}$, etc.

28. **Half-life.** The half-life of a radioactive substance is the time in which one-half of the initial number of nuclei disintegrates.

$$T_{1/2} = \frac{0.693}{\lambda}; \quad N = N_0 \left(\frac{1}{2} \right)^n$$

where n = number of half-lives in time $t = \frac{t}{T_{1/2}}$.

Its units are s, day, year, etc.

29. **Mean-life.** It may be defined as the ratio of the combined age of all the atoms to the total number of atoms present in the given sample.

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693} = 1.44 T_{1/2}$$

Its units are s, day, year, etc.

30. **Decay rate or activity of a sample.** It is the number of radioactive disintegrations taking place per second in a given sample.

$$R = \left| \frac{dN}{dt} \right| = \lambda N = \lambda N_0 e^{-\lambda t} \quad \text{or} \quad R = R_0 e^{-\lambda t}$$

31. **Becquerel.** It is the SI unit of activity and is defined as the decay rate of one disintegration per second.

$$1 \text{ becquerel} = 1 \text{ bq} = 1 \text{ decay per second.}$$

32. **Curie.** One curie is the decay rate of 3.7×10^{10} disintegrations per second.

$$1 \text{ Ci (curie)} = 3.70 \times 10^{10} \text{ disintegrations / s.}$$

33. **Rutherford.** One rutherford is the decay rate of 10^6 disintegrations per second.

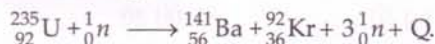
$$1 \text{ rd (rutherford)} = 10^6 \text{ disintegrations/s.}$$

34. **Natural radioactivity.** It is the phenomenon of the spontaneous emission of α -, β - or γ -radiations from the nuclei of naturally occurring isotopes.

35. **Artificial or induced radioactivity.** It is the phenomenon of inducing radioactivity in certain stable nuclei by bombarding them by suitable high energy particles.

36. **Nuclear reaction.** It is a reaction which involves the change of stable nucleus of one element into the nucleus of another element, by bombarding the former with suitable high energy particles.

37. **Nuclear fission.** It is the process in which a heavy nucleus ($A > 230$) when excited gets split up into two smaller nuclei of nearly comparable masses. For example,



38. **Thermal neutrons.** These are the slow moving neutrons of energy 0.0253 eV, corresponding to the velocities of 2200 ms^{-1} .

39. **Multiplication factor.** The multiplication factor of a fissionable mass is defined as the ratio of the number of neutrons present at the beginning of a particular generation to the number of neutrons present at the beginning of the previous generation.

If $k > 1$, the chain reaction *grows*.

If $k = 1$, the chain reaction remains *steady*.

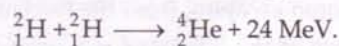
If $k < 1$, the chain reaction gradually *dies out*.

40. **Critical size and critical mass.** The size of the fissionable material for which multiplication factor is unity is called critical size and its mass is called critical mass of the material. The chain reaction in this case remains steady or sustained.

41. **Moderator.** Any substance which is used to slow down fast moving neutrons to thermal energies is called a moderator. The commonly used moderators are water, heavy water (D_2O) and graphite.

42. **Nuclear reactor.** It is a device in which a nuclear chain reaction is initiated, maintained and controlled. The reaction is controlled by using neutron absorbing materials like cadmium rods.

43. **Nuclear fusion.** It is the process of fusion of two smaller nuclei into a heavier nucleus with the liberation of large amount of energy. For example,



These reactions require the extreme conditions of temperature and pressure so that the reacting nuclei can overcome their electrostatic repulsion. For this reason, these reactions are called *thermonuclear reactions*.

COMPETITION SECTION

Semiconductor Devices and Digital Circuits

GLIMPSES

- Electronic devices.** Any device whose action is based on the controlled flow of electrons through it is called an *electronic device*. The branch of physics that deals with the study of these electronic devices is called *electronics*. The electronic devices are of *two* types :
 - Vacuum tubes
 - Solid-state electronic devices.
- Vacuum tubes.** These include vacuum diode, triode, tetrode, etc. In a vacuum tube, electrons obtained from a heated cathode are controlled by varying voltages between its different electrodes. These devices are bulky, consume high power, operate generally at high voltages, have limited life and low reliability.
- Solid-state electronic devices.** In such devices, the charge carriers flow through solid-state semiconductors. These devices include junction diodes, transistors and integrated circuits. These are small in size, consume low power, operate at low voltages, have long life and high reliability.
- Classification of solids on the basis of their resistivities.** Metals have low resistivity (10^{-2} to $10^{-8} \Omega\text{m}$), insulators have high resistivity ($>10^8 \Omega\text{m}$), while semiconductors have intermediate values (10^5 to $10^0 \Omega\text{m}$) of resistivity.
- Semiconductors.** They have much higher resistivity than metals. Their temperature coefficient of resistivity (α) is both negative and high. They have considerably lower number density of charge carriers than metals. Semi-conductors may be elemental (Si, Ge) and compound (GaAs, CdS, etc.)
- Energy bands in solids.** In an isolated atom, the electrons occupy well defined discrete energy levels. But due to interatomic interactions in a crystal, the electrons of the outer shells are forced to have energies different from those in isolated atoms. Each energy level splits into a number of energy levels forming a continuous band.

An enormously large number of energy levels closely spaced in a very small energy range constitute an *energy band*. The allowed energy bands are separated by regions in which energy levels cannot exist. These forbidden regions are called *band gaps* or *energy gaps*. The highest energy band occupied by the valence electrons is called the *valence band* and the next empty allowed band is called the *conduction band*.
- Fermi level.** The highest energy level filled with electrons at absolute zero is called Fermi level and the energy corresponding to the Fermi level is called Fermi energy.
- Distinction between metals, insulators and semiconductors on the basis of band theory.**
 - Metals.** In metals, either the conduction band is partially filled or the valence and conduction bands partly overlap.

Here $E_g = 0$.

This makes available a large number of free electrons for electric conduction. So metals have high conductivity or low resistivity.
 - Insulators.** Here the conduction band is empty and the valence band is filled. The forbidden energy gap is large ($E_g > 3 \text{ eV}$). Electrons cannot be excited from the valence band to the conduction band even by applying a strong electric field. Therefore, no electrical conduction is possible.

(iii) **Semiconductors.** The empty conduction band is separated from the filled valence band by a small energy gap ($E_g < 3 \text{ eV}$). Some electrons of the valence band easily get thermally excited to the conduction band and can conduct electricity. So semiconductors acquire small conductivity even at room temperature.

9. **Intrinsic semiconductors.** The pure semiconductors in which the electrical conductivity is totally governed by the electrons excited from the valence band to the conduction band and in which no impurity atoms are added to increase their conductivity are called intrinsic semiconductors and their conductivity is called intrinsic conductivity. Electrical conduction in pure semiconductors occurs by means of electron-hole pairs. In an intrinsic semiconductor,

$$n_e = n_h = n_i$$

where,

n_e = the free electron density in conduction band,

n_h = the hole density in valence band, and

n_i = the intrinsic carrier concentration.

10. **Doping.** The process of deliberate addition of a desirable impurity to a pure semiconductor so as to increase its conductivity is called doping. The impurity atoms added are called dopants. Dopants are of two types :

(i) **Pentavalent dopants** such as As, Sb and P. These are also called **donors**.

(ii) **Trivalent dopants** such as In, B and Al. These are also called **acceptors**.

11. **Extrinsic semiconductors.** A semiconductor doped with suitable impurity atoms so as to increase its conductivity is called an extrinsic semiconductor. Extrinsic semiconductors are of two types :

(i) ***n*-type semiconductors**, and

(ii) ***p*-type semiconductors**.

12. ***n*-type semiconductors.** The pentavalent impurity atoms are called donors because they donate electrons to the host crystal and the semiconductor doped with donors is called *n*-type semiconductor. In *n*-type semiconductors, electrons are the majority charge carriers and holes are the minority charge carriers. Thus

$$n_e \approx N_D > n_h$$

13. ***p*-type semiconductors.** The trivalent impurity atoms are called acceptors because they create holes which can accept electrons from the nearby bonds. A semiconductor doped with acceptor type impurities is called a *p*-type semiconductor. In *p*-type semiconductor, holes are the majority carriers and electrons are the minority charge carriers. Thus

$$n_h \approx N_A > n_e$$

In any semiconductor,

$$n_e n_h = n_i^2$$

Moreover, the material on the whole is electrically neutral.

14. **Holes.** The vacancy or absence of electron in the bond of a covalently bonded crystal is called a hole. A hole serves a positive charge carrier.

15. **Mobility.** The drift velocity acquired by a charge carrier in a unit electric field is called its electrical mobility and is denoted by μ .

$$\mu = \frac{v}{E}$$

SI unit of $\mu = \text{m}^2 \text{V}^{-1} \text{s}^{-1}$

The mobility of an electron in the conduction band is greater than that of the hole (or electron) in the valence band.

16. **Electrical conductivity of a semiconductor.** If a potential difference V is applied across a conductor of length l and area of cross-section A , then the total current I through it is given by

$$I = eA(n_e v_e + n_h v_h)$$

where n_e and n_h are the electron and hole densities, and v_e and v_h are their drift velocities, respectively. If μ_e and μ_h are the electron and hole mobilities, then the conductivity of the semiconductor will be

$$\sigma = e(n_e \mu_e + n_h \mu_h)$$

and the resistivity will be

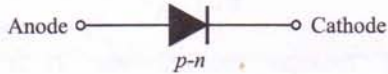
$$\rho = \frac{1}{e(n_e \mu_e + n_h \mu_h)}$$

The conductivity of an intrinsic semiconductor increases exponentially with temperature as

$$\sigma = \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right).$$

17. ***p-n junction*.** It is a single crystal of Ge or Si doped in such a manner that one half portion of it acts as *p*-type semiconductor and other half functions as *n*-type semiconductor. As soon as a

p - n junction is formed, the holes from the p -region diffuse into the n -region and electrons from n -region diffuse into p -region. This results in the development of **potential barrier** V_B across the junction which opposes the further diffusion of electrons and holes through the junction. The small region in the vicinity of the junction which is depleted of free charge carriers and has only immobile ions is called the **depletion region**.



Symbol for a p - n junction diode.

18. **Forward and reverse biasing of a p - n junction.** If the positive terminal of a battery is connected to the p -side and the negative terminal to the n -side, then the p - n junction is said to be **forward biased**. Both electrons and holes move towards the junction. A current, called forward current, flows across the junction. Thus a p - n junction offers a low resistance when it is forward biased.

If the positive terminal of a battery is connected to the n -side and negative terminal to the p -side, then p - n junction is said to be **reverse biased**. The majority charge carriers move away from the junction. The potential barrier offers high resistance during the reverse bias. However, due to the minority charge carriers a small current, called reverse or leakage current, flows in the opposite direction. Thus a junction diode has almost a unidirectional flow of current.

The forward bias current is large (mA) while the reverse bias current is small (μ A) in a p - n junction diode.

19. **Dynamic resistance.** The dynamic or ac resistance of a diode is the ratio of small change in applied voltage ΔV to the corresponding change in current ΔI . It is given by

$$r_d = \frac{\Delta V}{\Delta I}$$

20. **Characteristic of a p - n junction diode.** A graph showing the variation of current flowing through a p - n junction with the voltage applied across it (both when it is forward and reverse biased) is called the **voltage-current** or **V-I characteristic** of a p - n junction.

21. **Rectification.** The process of converting a.c. into d.c. is called rectification and the device used for this purpose is called a rectifier.
22. **Junction diode as a rectifier.** A junction diode conducts when forward biased and does not conduct when reverse biased. This unidirectional characteristic of the diode enables it to be used as a rectifier. A half-wave rectifier uses only a single diode while a full wave rectifier uses two diodes.
23. **Zener diode.** A junction diode specially designed to work only in the reverse breakdown region continuously is called a zener diode. The voltage drop across it is practically independent of current through it. So it can be used as a **voltage regulator**.
24. **Photodiode.** It is a junction diode made from a photosensitive semiconducting material in such a way that light can fall on its junction. It is operated in a reverse bias condition. The photon excitation results in a change of reverse saturation current which helps us to measure light intensity.
25. **Light emitting diode (LED).** It is a forward biased p - n junction which spontaneously converts the biasing electrical energy into optical energy, like infra-red and visible light. It is made from a translucent semiconductor like GaAs or InP. LEDs are used in hand calculators, cash registers, digital clocks, etc.
26. **Solar cell.** It is a junction diode used to convert solar energy into electrical energy. It is based on photovoltaic effect (generation of voltage due to bombardment of light photons). The materials used for solar cells are Si and GaAs.
27. **Junction transistor.** It is a three terminal solid state device obtained by growing either a narrow section of p -type crystal between two relatively thicker sections of n -type crystals or a narrow section of n -type crystal between two thicker sections of p -type crystals. The first type is called n - p - n transistor and the second type is called p - n - p transistor. Each type of transistor has three components :

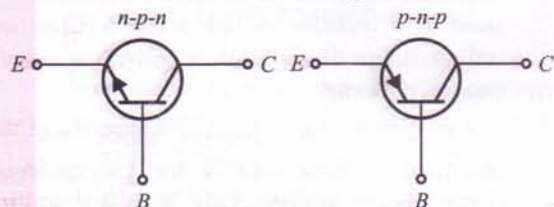
Emitter (E), Collector (C) and Base (B).

The base is very thin and lightly doped. The emitter is always forward biased and the collector is always reverse biased.

The emitter supplies a large number of majority charge carriers for the flow of current through the transistor.

The base controls the flow of majority charge carriers from emitter to cathode.

The collector collects the majority charge carriers for the transistor operation.



Circuit symbol of (a) $n-p-n$ (b) $p-n-p$ transistors.

28. **Action of transistor.** When the emitter-base junction is forward biased, majority charge carriers are pushed from emitter to base. Few electrons and holes recombine in the base region. Most of the majority charge carriers cross over to the reverse biased collector. Collector current I_C is always slightly less than the emitter current.

$$I_E = I_C + I_B, \text{ where } I_B \text{ is base current.}$$

29. **Three configurations of a transistor.** A transistor can be used in one of the following three configurations :

- (i) Common-base (CB) circuit
- (ii) Common-emitter (CE) circuit
- (iii) Common-collector (CC) circuit.

30. **Common-emitter characteristics of a transistor.** These are the graphs between appropriate voltages and currents for a transistor when its emitter is taken as the common terminal and grounded (zero potential), base is the input terminal and collector is the output terminal. Three types of characteristic curves are studied :

- (i) **Input characteristic.** It is the graph showing the variation of base current I_B with base-emitter voltage V_{BE} at constant collector-emitter voltage V_{CE} .

The input resistance r_i of a transistor in CE configuration is defined as the ratio of the small change in the base-emitter voltage to the corresponding small change in base current, when the collector-emitter voltage is kept fixed.

$$r_i = \left[\frac{\Delta V_{BE}}{\Delta I_B} \right]_{V_{CE} = \text{constant}}$$

- (ii) **Output characteristic.** It is the graph that shows the variation of collector current I_C with the collector-emitter voltage V_{CE} at constant base-current I_B .

The output resistance r_o of a transistor in CE configuration is defined as the ratio of the small change in the collector-emitter voltage to the corresponding change in the collector-current when the base current is kept constant.

$$r_o = \left[\frac{\Delta V_{CE}}{\Delta I_C} \right]_{I_B = \text{constant}}$$

- (iii) **Transfer characteristic.** It is a graph showing the variation of collector current I_C with base current I_B at constant collector-emitter voltage V_{CE} .

31. **Current gains of a transistor.** Usually two current gains are defined :

- (i) **Common base current amplification factor or ac current gain α .** It is the ratio of the small change in the collector current to the small change in the emitter current when the collector-base voltage is kept constant.

$$\alpha = \left[\frac{\Delta I_C}{\Delta I_E} \right]_{V_{CB} = \text{constant}}$$

- (ii) **Common emitter current amplification factor or ac current gain β .** It is the ratio of the small change in the collector current to the small change in the base current when the collector-emitter voltage is kept constant.

$$\beta = \left[\frac{\Delta I_C}{\Delta I_B} \right]_{V_{CE} = \text{constant}}$$

32. **Relations between α and β .** The current gains α and β are related as

$$\alpha = \frac{\beta}{1 + \beta} \quad \text{and} \quad \beta = \frac{\alpha}{1 - \alpha}$$

33. **Three states of a transistor.** A transistor in a circuit can be in one of the three conditions :

- (i) If the emitter-base junction is not forward-biased or if the bias is insufficient to start forward current across the junction while collector-base junction is reverse-biased, then the transistor is said to be in **cutoff state**.
- (ii) If the emitter-base junction is forward-biased and the collector base junction is reverse biased, a collector current flows in linear proportion with the base current. The transistor is said to be in the **active state**.

(iii) If the emitter-base junction is heavily forward biased and the collector current produces such large drop across the load resistor that the base-collector junction also gets forward biased, then the transistor is said to be in **saturation state**.

34. **Transistor as a switch.** A transistor can be used as a switch if it is biased in its cut off and saturation states only. The active state is forbidden when it is used as a switch.

35. **Transistor as an amplifier.** An amplifier is a circuit which is used for increasing the voltage, current or power of alternating form. A transistor operates as an amplifier when its emitter-base junction is forward biased while collector-base junction is reverse-biased. Transistor biased in this way is said to be in active state.

36. **Common base transistor amplifier.** (i) **d.c. current gain.** It is defined as the ratio of collector current to the emitter current.

$$\alpha_{dc} = \frac{I_C}{I_E}$$

(ii) **a.c. current gain.** It is defined as the ratio of the small change in collector current to the small change in emitter current.

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$$

(iii) **Voltage gain.** It is defined as the ratio of the small change in output voltage to the small change in input voltage.

$$A_v = \frac{-\Delta V_{CB}}{\Delta V_{EB}} = \frac{\Delta I_C \times R_o}{\Delta I_E \times R_i}$$

or
$$A_v = \alpha_{ac} \cdot \frac{R_o}{R_i} = A_i \times A_r$$

(iv) **Power gain.** It is defined as the ratio of output power to input power.

$$A_p = \frac{\Delta V_{CB}}{\Delta V_{EB}} \times \frac{\Delta I_C}{\Delta I_E} = A_v \cdot A_i = A_v \cdot \alpha_{ac} = \alpha_{ac}^2 \cdot \frac{R_o}{R_i}$$

37. **Common emitter transistor amplifier.**

(i) **d.c. current gain.** It is defined as

$$\beta_{dc} = \frac{I_C}{I_B}$$

(ii) **a.c. current gain.** It is defined as

$$\beta_{ac} \text{ or } A_i = \frac{\Delta I_C}{\Delta I_B}$$

(iii) **Voltage gain.** It is defined as

$$A_v = \frac{\Delta V_{CE}}{\Delta V_{BE}} = \beta_{ac} \cdot \frac{R_o}{R_i} = A_i \cdot A_r$$

(iv) **Power gain.** It is defined as

$$A_p = \frac{\text{Output power}}{\text{Input power}} = \frac{\Delta V_{CE}}{\Delta V_{BE}} \cdot \frac{\Delta I_C}{\Delta I_B} = A_v \cdot \beta_{ac} = \beta_{ac}^2 \cdot \frac{R_o}{R_i}$$

(v) **Transconductance (g_m).** It is defined as the ratio of small change in collector current (ΔI_C) to the small change in input base-emitter voltage.

$$g_m = \frac{\Delta I_C}{\Delta V_{BE}}$$

Its unit is Ω^{-1} or siemen (S). It depends on the geometry, doping levels and biasing of the transistor.

38. **Oscillator.** An oscillator is an electronic device which produces electric oscillations of constant frequency and amplitude. An LC-circuit is the basic oscillatory circuit. But its oscillations are damped. When a transistor is used an amplifier with positive feedback, it acts as an oscillator. Here the oscillations are set up in the tank or LC-circuit and feed-back energy is supplied in the correct phase by an inductance L' which is inductively coupled with the inductance L of the tank circuit.

The frequency of the oscillations in the tank circuit is

$$f = \frac{1}{2\pi\sqrt{LC}}$$

39. **Analog circuits.** These circuits process a signal (current or voltage) in the form of continuous, time-varying voltage or current. Such signals are called continuous or analog signals.

40. **Digital circuits.** These circuits, instead of using continuous signals, make use of discrete two-level or binary signals. Such signals are called digital signals. Here only two values (0 and 1) of the input and output voltage are permissible.

41. **Logic gate.** A logic gate is a digital circuit that has one or more inputs but only one output. It follows a logical relationship between input and output voltages.

42. **Truth table.** This table shows all possible input combinations and the corresponding outputs for a logic gate.
43. **Boolean expression.** It is a shorthand method to describe the function of a logic gate in the form of an equation or an expression. It also relates the all possible combinations of the inputs of a logic gate to the corresponding outputs.
44. **Positive and negative logic.** If in a system, the higher voltage level represents 1 and the lower voltage level represents 0, the system is called a positive logic. If the higher voltage level represents 0 and the lower voltage level represents 1, then the system is called a negative logic.
45. **OR gate.** It is a digital circuit having two or more inputs but only one output. It gives a high output if either input A or B or both are high (1) otherwise it gives low output (0). It is described by Boolean expression :

$$A + B = Y$$

Logic symbol of
OR gate

Truth table of
OR gate



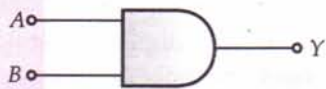
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

46. **AND gate.** It is a digital circuit having two or more inputs but only one output. It gives a high output (1) if inputs A and B are both high (1) otherwise it gives low output (0). It is described by the Boolean expression :

$$A \cdot B = Y$$

Logic symbol of
AND gate

Truth table of
AND gate



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

47. **NOT gate.** It is a digital circuit having only one input and one output. It gives a high output (1) if the input A is low (0) and vice versa. It is described by the Boolean expression,

$$\bar{A} = Y$$

Logic symbol of
NOT gate

Truth table of
NOT gate



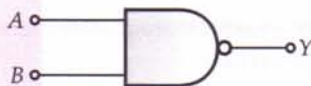
A	B
0	1
1	0

48. **NAND gate.** It is obtained by connecting the output of an AND gate to the input of a NOT gate. Its input is high if both inputs A and B are not high. It is described by the Boolean expression,

$$\overline{A \cdot B} = Y$$

Logic symbol of
NAND gate

Truth table of
NAND gate



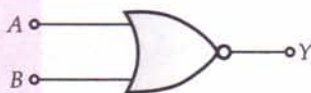
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

49. **NOR gate.** It is obtained by connecting the output of an OR gate to the input of a NOT gate. Its output is high if neither input A nor input B is high. It is described by the Boolean expression,

$$\overline{A + B} = Y$$

Logic symbol of
NOR gate

Truth table of
NOR gate



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

50. **XOR or exclusive OR gate.** The XOR gate gives a high output if either input A or B is high but not when both A and B are high or low. It can be obtained by using a combination of two NOT gates, two AND gates and one OR gate. It is described by Boolean expression : $Y = \bar{A}B + A\bar{B}$

The XOR gate is also known as *difference gate* because its output is high when the inputs are different.

Logic symbol of
XOR gate



Truth table of
XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

51. **Integrated circuits (ICs).** A miniature electronic circuit, consisting of many passive components like R and C and active components like diode and transistor, fabricated within a single semiconductor chip is called an integrated circuit. ICs are obtained by a complex procedure involving diffusion, oxidation, photolithography, metallisation, etc. These circuits have revolutionised the electronic industry.

COMPETITION SECTION

Communication Systems

GLIMPSES

1. **Communication.** It is the process by which information is transferred faithfully from one point to the other point in an intelligible form.
2. **Message signal.** It is defined as the single-valued function of time (that conveys the information) and which, at every instant of time takes a unique value. Signals can be *analog* or *digital*.
3. **Analog signal.** A signal in which current or voltage varies continuously with time is called analog signal.
4. **Digital signal.** A signal in which current or voltage can take only two discrete values is called a digital signal.
5. **Communication system.** The setup used to transmit information from one point to another is called communication system. A communication system mainly consists of (i) *transmitter*, (ii) *communication channel* and (iii) *receiver*.
6. **Analog communication.** It makes use of analog electronic circuit and analog signal in such a way that the output voltage varies continuously in accordance with the input voltage. For example, telegraphy, telex, TV network, etc.
7. **Digital communication.** It makes use of an electronic circuit that can handle only digital signals. For example, FAX, E-mail, etc.
8. **Transmitter.** It is a setup that transmits the message to the receiving end through a communication channel. It modifies the message signal in a form suitable for transmission and then transmits it. It mainly consists of (i) *message signal generator*, (ii) *modulator* and (iii) *antenna*.
9. **Bandwidth.** It refers to the range over which the frequencies in a signal vary. It is equal to the difference between the highest and lowest frequencies present in a signal.
10. **Baseband.** It refers to the band of frequencies of the original signal as is produced by the source of information.
11. **Transducer.** Any device which converts energy from one form to another is called a transducer. An electrical transducer converts some physical variations in pressure, displacement, temperature, etc. into corresponding variations in an electrical signal.
12. **Noise.** These are unwanted signals which get mixed up with the information signal during its propagation through the communication channel.
13. **Attenuation.** It is the loss of strength of a signal during its propagation through the communication channel.
14. **Amplification.** It is the process of increasing the amplitude and hence the strength of an electrical signal by using an electronic circuit (consisting of atleast one transistor).
15. **Range.** It is defined as the largest distance between the source and the destination upto which a signal can be received with sufficient strength.
16. **Repeater.** A repeater is a combination of a transmitter, an amplifier and a receiver which picks-up a signal from the transmitter, amplifies and retransmits it to the receiver sometimes with a change of carrier frequency.

17. **Nyquist sampling theorem.** An analog signal can be perfectly reconstructed from its samples without any loss of information if it is sampled atleast twice the rate of the maximum frequency component or the bandwidth of the signal. This is called sampling theorem.

$$f_s > 2 f_m$$

where f_s is the sampling frequency and f_m is the maximum frequency of the signal.

18. **Modulation.** A high frequency carrier wave is used to carry the audio signal to large distances. Modulation is the process of changing some characteristic (amplitude, frequency or phase) of high frequency carrier wave in accordance with the instantaneous value of the low frequency audio signal called the modulating signal. Correspondingly, we have three types of modulation : (i) amplitude modulation (ii) frequency modulation and (iii) phase modulation.

19. **Need for modulation.** The low frequency information signals (i) radiated directly from an antenna die out after travelling some distance, (ii) require antenna of impractically large length and (iii) transmitted from different stations get inseparably mixed up. Due to the above reasons, low frequency information signals need to be modulated by a high frequency carrier wave.

20. **Amplitude modulation.** It is the process in which the amplitude of the high frequency carrier wave changes in accordance with the instantaneous value of the modulating signal. Here a voltage proportional to the amplitude of the modulating signal gets added to the carrier amplitude.

21. **Modulation factor.** It is defined as the ratio of the change in the amplitude of carrier wave to the amplitude of original carrier wave. It is also known as degree of modulation or modulation index. If A_m and A_c are the amplitudes of modulating and carrier signals respectively, then the modulation factor is given by

$$\begin{aligned} \mu &= \frac{\text{Change in amplitude of carrier wave}}{\text{Amplitude of original carrier wave}} \\ &= \frac{A_m}{A_c} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \end{aligned}$$

where $A_{\max}$ and $A_{\min}$ are the maximum and minimum voltages of AM wave.

22. **Expression for amplitude modulated wave.**

Let modulating signal, $m(t) = A_m \sin \omega_m t$

And carrier signal, $c(t) = A_c \sin \omega_c t$

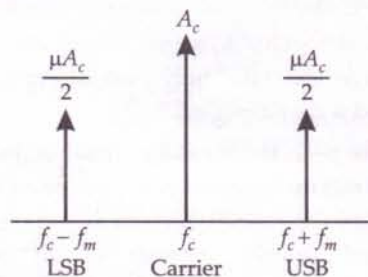
The instantaneous voltage of the A.M. wave is

$$c_m(t) = A_c \sin \omega_c t + \frac{\mu A_c}{2} [\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t] \quad \dots(i)$$

23. **Sideband frequencies and bandwidth of A.M. wave.** The equation (i) shows that the A.M. wave is the sum of three sinusoidal waves :

- One the original carrier wave of amplitude A_c and frequency $f_c = \omega_c / 2\pi$.
- The second wave of amplitude $\mu A_c / 2$ and of frequency $(f_c + f_m)$. The sum of carrier frequency and modulating frequency $(f_c + f_m)$ is called *upper side band frequency* (USB).
- The third wave of amplitude $\mu A_c / 2$ and frequency $(f_c - f_m)$. The difference of carrier frequency and modulating frequency $(f_c - f_m)$ is called *lower side band frequency* (LSB).

The frequency spectrum of an A.M. is shown below :



The difference between the highest and the lowest frequencies present in the A.M. wave is called its bandwidth.

$$\text{Bandwidth} = (f_c + f_m) - (f_c - f_m) = 2 f_m$$

Hence bandwidth of the A.M. wave is twice the frequency of the modulating signal.

24. **Frequency modulation.** It is the process of changing the frequency of the carrier voltage in accordance with the instantaneous value of the modulating voltage. The original frequency of the carrier signal is called *resting frequency* or *centre frequency* (f_c). The amount by which the frequency of the carrier wave changes or shifts above or below the resting frequency is called *frequency deviation* (Δf).

Clearly, $\Delta f \propto m(t)$

25. **Carrier swing.** The total variation in frequency of F.M. wave from the lowest to the highest is called carrier swing (CS).

Carrier swing

$$= 2 \times \text{Frequency deviation in centre frequency}$$

$$CS = 2 \Delta f$$

26. **Modulation index in F.M.** It is defined as the ratio of frequency deviation to the modulating frequency. So

$$\mu_f = \frac{\text{Frequency deviation}}{\text{Modulating frequency}}$$

$$= \frac{\Delta f}{f_m}$$

27. **Bandwidth in F.M.** If the various sidebands are separated by the modulating frequency f_m , then

$$\text{Bandwidth} = 2n \times \text{modulating frequency}$$

$$= 2n \times f_m$$

where n is the number of significant bands present in the F.M. wave.

28. **Phase modulation.** It is the process of varying the phase angle of the carrier voltage in accordance with the instantaneous value of the modulating voltage.

29. **Pulse modulation.** It is a process in which some parameter of a pulse train is varied in accordance with the message signal or the modulating signal.

30. **Different types of pulse modulation :**

(i) **Pulse amplitude modulation (PAM).** It is defined as the process in which amplitudes of the regularly spaced rectangular pulses vary in direct proportion to the instantaneous values of continuous message signal.

(ii) **Pulse duration modulation (PDM).** Here the starting time and amplitude of each pulse are constant but the width or duration of each pulse is made proportional to the instantaneous value of the analog signal.

(iii) **Pulse position modulation (PPM).** Here the position of a pulse relative to its unmodulated time of occurrence is varied in accordance with the instantaneous value of the message signal.

(iv) **Pulse code modulation (PCM).** It is a digital process in which the message signal is sampled and the amplitude of each sample is rounded off to the nearest value of a finite set of allowable values and rounded values are coded. The signals are transmitted as a binary code.

31. **Antenna.** An antenna is basically a small length of a conductor that is used to radiate or receive electromagnetic waves. It acts as a conversion device. At the transmitting end, it transforms high frequency current into electromagnetic waves. At the receiving end, it transforms electromagnetic waves into electrical signal that is fed to the input of the receiver. A dipole antenna is omnidirectional, whereas the dish antenna is highly directional. The length of a dipole antenna is usually $\lambda/2$.

32. **Receiver.** Its function is to recover the original message or data from the modulated signal after its propagation through the communication channel. It separates the modulating signal from the carrier wave. The process is called demodulation.

33. **Demodulation.** The process of recovering the message signal from the modulated wave is known as modulation or detection. A junction diode can be used as a detector or modulator.

34. **Modem.** A modem is a device that connects a transmitting digital device to a receiving digital device through telephone line. The name modem is a contraction of the terms modulator and demodulator. In the transmitting mode, a modem acts as a modulator and in the receiving mode, it acts as a demodulator.

35. **Communication channels.** The physical path between the transmitter and receiver is called the communication channel or the transmission medium.

The transmission media can be broadly divided into two categories :

(i) **Guided media.** Such media are used for point-to-point communication. For example, twisted pair, coaxial cable and optical fibre.

The quality of transmission in guided media is determined mainly by the nature of the media.

(ii) **Unguided media.** Free space is an example of unguided medium. Here the frequency band of the transmitted signal is more important than the medium in determining the quality of transmission.

Depending on the nature of the communication channel, the communication is classified broadly into three categories :

- (a) Line communication
- (b) Optical communication
- (c) Space communication

36. **Bandwidth of a communication channel.** It is defined as the difference between the highest and the lowest frequencies that communication channel allows to pass through it. It must be equal to or greater than the bandwidth of the information signal.

Number of channels

$$= \frac{\text{Total bandwidth of channel}}{\text{Bandwidth needed per channel}}$$

37. **Space communication.** Here electromagnetic waves of different frequencies are used to carry information through the physical space acting as the transmission medium.

It is of three types :

- (i) **Ground wave propagation.** A radiowave that can travel directly from one point to another following the surface of the earth is called a ground wave. As the ground wave passes over this surface, it is attenuated due to the absorption of its energy by the earth. This absorption increases with the increase in frequency. Hence ground wave propagation is possible at low frequencies (500 kHz to 1500 kHz). This region of AM band is called *medium wave band*.
- (ii) **Space wave propagation.** If a radiowave transmitted from an antenna, travelling in a straight line, directly reaches the receiving antenna, it is called a space wave. The space wave propagation is also called tropospheric or line of sight propagation. Television frequencies (100 – 220 MHz) are transmitted through this mode.
- (iii) **Sky wave propagation.** A radiowave directed towards the sky and reflected by

the ionosphere towards the desired location of the earth is called a sky wave. Radiowaves of frequencies 2 – 20 MHz can be reflected by the ionosphere. This region of the AM band is called *short wave band*.

38. **Critical frequency.** It is the highest frequency of radiowave which when sent straight (normally) towards the given layer of ionosphere gets reflected from ionosphere and returns to the earth. Above this frequency, the radiowave will penetrate the ionosphere and is not reflected back.

$$f_c = 9 (N_{\max})^{1/2}$$

where $N_{\max}$ = the maximum electron density of the ionosphere.

39. **Skip distance.** It is defined as the smallest distance measured from a transmitter, measured along the earth's surface to a point where the sky wave of fixed frequency but less than critical frequency returns to the earth after reflection from the ionosphere.

40. **Range of TV transmission.** The signals transmitted from antenna of height h can be directly received by a receiver on the surface of the earth within the range given by

$$d = \sqrt{2hR}, \quad R \gg h$$

41. **Range of line-of-sight distance between two antennas.** The maximum line-of-sight distance d_M between transmitting and receiving antennas of heights h_T and h_R is given by

$$d_M = d_T + d_R = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where R is the radius of the earth.

42. **Satellite communication.** It is the mode of communication of a signal between transmitter and receiver through a satellite. A beam of modulated microwave is sent directly towards the satellite from a transmitter on the earth. The signals are received, processed, amplified and retransmitted by a device called *transponder* fitted in the satellite. The signals received on the earth are amplified and retransmitted by a receiver.

43. **Transponder.** It is a device fitted on a satellite which receives the signal and retransmits it after amplification.

44. **Remote sensing.** It is the technique of obtaining information about an object, area or phenomenon, acquired by a sensor that is not in direct contact with the target of investigation. A satellite equipped with suitable sensors to acquire data is placed in an orbit around the earth at any height having a period of revolution. It takes photographs or collects any other information desired and transmits it back to an earth station. This is known as remote sensing.
45. **Line communication.** A communication system involving point-to-point contact between the transmitter and receiver is called line communication. This mode of communication requires use of a guided medium such as a parallel-wire line, twisted pair or coaxial cable.
46. **Parallel-wire line.** It consists of two conducting wires running parallel to each other inside a protective insulating PVC coating. This system is very lossy and suffers from electromagnetic interference from other sources.
47. **Twisted pair.** It consists of two insulated copper wires twisted around each other almost regularly. It can transmit both analog and digital signals and is less prone to electrical disturbances.
48. **Coaxial cable.** It consists of a hollow cylindrical conductor, surrounding a coaxial single inner conductor. The inner conductor is a copper wire surrounded by a low dielectric insulator and the outer conductor is a copper mesh covered with PVC. The outer conductor is normally grounded and thus provides an electrical shield to the signals carried by the central conductor. Coaxial cables are used for long distance high frequency transmission, local area computer networks for high-speed data transmission and as input / output channels for the computer systems.
49. **Optical communication.** It is a mode of communication in which a light beam carries information from one point to another by using a guided medium like optical fibre. This method has *two* main advantages :
 - (i) It can be used as a high bandwidth carrier.
 - (ii) It can carry large amount of information.
50. **Internet.** It is a global network of computers linked by high speed data lines and wireless systems. It allows communication and sharing of information between any two or more computers.
51. **Email : Electronic mail.** It is a message sent and received through a computer network.
52. **File Transfer.** An FTP (File Transfer Protocol) permits the transfer of files/software from one computer to another connected to the internet.
53. **WWW : World Wide Web.** It is a set of protocols that allows us to access any document on the internet.
54. **E-commerce.** It is the collection of tools and practices involving internet technologies that allow a company to create, maintain and optimise business relations with consumers and other businesses. It permits online banking and shopping.
55. **Chat.** It is the real time conversation among people with common interests through the typed messages on the net.
56. **FAX : Facsimile telegraphy.** The electronic reproduction of a document at a distant place is called Facsimile telegraphy or fax.

In order to send a document through fax, the following three functions are performed :
 - (i) At the sending end, the original document is read by an optical scanner.
 - (ii) The black and white dots of the document are encoded and compressed for transmission and reception via a telephone line.
 - (iii) At the receiving end, the received data is decompressed and processed by a printer to reproduce the original document.
57. **Mobile Telephony.** Instead of a single high-power transmitter for an entire service area, numerous lower power transmitters (*base stations*) are set up, each covering a fraction of that service area, called a *cell*. As a mobile receiver moves from one cell to another, the mobile user is handed over to the new cell's base station through a switching office called *mobile telephone switching office (MTSO)*.